The Substratum Construction: Reconstruction, the Substratum Gauge Group, and the QM-GR Synthesis

Observer geometry and physical carrier. SM §4.1 distinguishes two claims. First, finite equivariant projection is classical and exact: cubic symmetry is inherited by the projected Markov part and every memory kernel, forbidding quadratic anisotropy in the scalar symbol — so whatever nondegenerate quadratic structure survives is isotropic — while allowing cubic anisotropy at quartic order. The particular normalized nearest-neighbor operator $A/(2d)$ additionally requires an **observer-level** center-free kernel; microscopic center independence alone does not imply that after correlated hidden variables are marginalized. Second, the six signed simple-cubic links and their exact $3 \oplus 2 \oplus 1$ representation are unconditional geometry, but identifying that link space with the complete physical carrier is **H-link**; the single-copy clause gives $K = 6$, and the custodial condensate-stabilizer reading is **H-cust**. Accordingly, any downstream statement here that consumes $K = 6$, $N_f = 6$, the physical SM gauge carrier, or a fixed three-sector count inherits those named bridge conditions unless it is explicitly referring only to the six geometric links.

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Abstract

Quantum mechanics and general relativity are conventionally treated as two fundamental theories awaiting unification. This paper develops the alternative that they are two projections of a single finite deterministic construction, joined by a third projection that produces the arrow of time. The underlying object is the bijection (S, φ) on a finite configuration space. Its visible-sector reduction has the exact finite stochastic/quantum-representation equivalence established in [Main]: finite observable laws, finite reversible realizations, and fixed-basis unitary/Born representations coincide, while genuine history readback forces hidden predictive memory. Its spatial coupling graph supplies finite causal cones; equality with the complete standard local quantum instrument theory is the narrower operational-lifting statement explicitly scoped in [Main]. The boundary-thermodynamic projection yields the gravitational description developed in [GR], and the lattice dynamics yields the Standard-Model structure developed in [SM].

The two substratum-level results that make these emergences one construction rather than independent applications are the reconstruction theorem (Theo-

rem 23) and the substratum gauge group (Theorem 24). Theorem 23 reconstructs the **local propagating lattice/gauge residue** from its stated empirical inputs and structural assumptions, conditional on the named modeling and completeness premises; Bell-inclusive existence additionally requires H-Bell and Bell-inclusive uniqueness is not established. Theorem 24 identifies the observables-preserving kernel of the dilation-datum/local reconstruction, conditional on Lemma 24.1's semigroup-transfer completeness step. The synthesis claim is therefore structural rather than rhetorical: the same finite deterministic object supports the visible stochastic/quantum projection, the boundary-thermodynamic gravitational projection, and the lattice gauge projection. The traditional QM–GR "unification problem" is correspondingly recast as a category error between two projections of one object, while the theorem statuses and remaining conditional steps are kept explicit.

1. Introduction

The standard physical picture of the world contains two fundamental theories. Quantum mechanics describes the microscopic regime in terms of unitary evolution on a Hilbert space, superposition, and Born-rule probabilities. General relativity describes the gravitational regime in terms of a deterministic, classical, dynamical metric obeying Einstein's equations. The two are mathematically and conceptually incompatible. Quantum mechanics has no preferred temporal foliation and no observer-independent state; general relativity has both. Quantum mechanics treats measurement as a primitive operation; general relativity treats it as a process within the dynamics. Quantum mechanics is linear; general relativity is not. Every attempt to unify them — to quantize gravity, to geometrize quantum mechanics, to merge the two into a deeper underlying theory — has produced either inconsistencies, ambiguities, or empirically unfalsifiable conjectures. The unification problem has resisted solution for nearly a century, and the most active current programs (loop quantum gravity, string theory, causal dynamical triangulations, asymptotic safety) have produced neither a quantitative empirical confirmation nor a clean conceptual resolution.

This paper argues that the QM-GR incompatibility is not a technical problem to be solved by finding the right mathematical framework. It is a category error: an attempt to merge two descriptions that turn out to be projections of the same underlying object viewed from different perspectives, rather than two competing accounts of the same regime. The framework developed in this four-paper synthesis to which this paper belongs — [Main], [SM], [GR], and the present paper — provides the underlying object and the projection map. The underlying object is a finite deterministic bijection (S, φ) on a configuration space partitioned into a visible sector accessible to an embedded observer and a hidden sector beyond the observer's causal reach. The projection from the underlying object to the observer's reduced description is the trace-out over the hidden sector. [Main] proves the exact finite observable-law equivalence $S \iff D \iff Q_{\mathrm{fb}}$: finite stochastic laws, finite reversible realizations, and

fixed-basis unitary/Born representations coincide. The representation itself is universal; the memory-bearing OI sector is singled out instead by history readback, which forces hidden predictive memory. For spatial systems the same construction supplies graph causal cones for finite-range couplings. What remains conditional on the quantum side is the strongest operational reading — one common standard quantum instrument/composite theory for the whole coherent intervention family. Bell is separate: the response-complete deterministic layer keeps measurement independence and reproduces quantum Bell violation only on the ontically parameter-dependent branch of [Main §3.3], with operational no-signaling. The projection from the same underlying object to general relativity is the thermodynamic limit at the partition boundary, formalized in [GR] as the cosmological-horizon application: applied to the cosmological horizon as a causal partition, the framework relates the action scale to the discreteness scale through the calibration $\hbar = c^3\epsilon^2/(4G\sigma)$, which fixes only $\epsilon/\sqrt{\sigma} = 2\,l_p$ for an entropy σ per boundary mode, and recovers the Bekenstein-Hawking entropy with the $1/4$ coefficient as a consistency relation of that calibration, both conditional on H-slope [GR §3.2], and proposes a dissolution of the cosmological constant problem conditional on the covariant coupling step [GR §6.2]. The Standard Model emerges from the same bijection on a cubic lattice with the wave equation as substratum dynamics: the cubic lattice is selected by the observed gauge group (the SU(3) factor requires the cubic crystal system), and — given that lattice — three generations (physical reading conditional on H-spin', [SM §4.7]; chirality on H-χ', [SM §4.8]), the Higgs as composite, and the $(3, 2, 1)$ gauge structure follow as forced consequences [SM]; strong CP is open there (H-top, H-det). The framework's hierarchical structural realism and its relationship to other unification programs are developed in the fifth core paper [Structure].

The present paper develops the substratum-level results that make these three emergences — emergent QM in [Main], the Standard Model derivation in [SM], and the gravitational sector in [GR] — a single construction rather than three separate applications of the same formalism. The two technical results that do this work are the *reconstruction theorem* (§3) and the *substratum gauge group* (§4). The reconstruction theorem shows — under its stated conditions (M1-T; C2-mixing; Lemma 24.1's completeness step, together with the Stage-2 structural inputs) — that observed physics fixes the **local propagating lattice/gauge residue** up to the stated gauge freedoms. Bell violation enters separately through M1-B: a common measurement-independent deterministic completion exists only conditional on H-Bell, and no theorem here makes the Bell-inclusive completion unique. The substratum gauge group identifies the kernel of this inverse map — four families of transformations of (S, φ), one of them (alphabet change) under the alphabet-size hypothesis of [SM §2.7], that exhaust — conditional on Lemma 24.1's completeness step — all observables-preserving operations of the dilation-datum substratum (§4) — and shows that the Standard Model gauge group is its shadow under the trace-out. The combination of these two results turns the framework's three derivations into a single object: the SM gauge group is not chosen from a landscape, the gravitational

thermodynamics are not appended to a quantum description, and the emergent QM is not postulated as a starting point. All three emerge from the same (S, φ), and the substratum gauge group is the symmetry that makes the relationship between them exact rather than approximate.

The synthesis claim that follows from these results — developed in §5 — is that quantum mechanics and general relativity are not two theories awaiting unification but two projections of the same construction, viewed at different levels of description. Quantum mechanics lives at the level of the visible-sector trace-out: the embedded observer sees the bijection through the hole of finite causal access, and the resulting compressed description admits the universal unitary quantum representation. General relativity lives at the level of the boundary thermodynamics: the same causal partition that produces, at the visible-sector level, the reduced description admitting the quantum representation produces classical horizon thermodynamics at the boundary level, and Jacobson's thermodynamic argument promotes this to Einstein's equations. The two descriptions are not in tension because they refer to different objects in the construction — not different *aspects of the same regime*, which would invite unification, but different *projections of the same substratum*, which makes unification a category error. The substratum gauge group makes this precise: at the level of (S, φ), there is no quantum and no classical, no metric and no wave function; there is only the bijection and its symmetries. The trace-out at the visible-sector level produces one set of derived structures (the wave function, the SM gauge group, particle content), and the thermodynamic limit at the boundary level produces another (the metric, $\hbar$, the BH entropy, the dynamical dark energy), and the framework provides the construction that makes both of these descriptions exact projections of the same object. The same construction produces a third projection (§5.4) that resolves the arrow-of-time question by descending the observer's emergent arrows — cosmological, thermodynamic, memory, measurement — from the horizon-growth clock and the C1–C4 observer-selection theorem of [Main, §4.6], answering the Ellis objection to bijective substrates without adding a structural commitment.

The paper is organized as follows. §1.1 states the framework's primary metaphysical commitment and its relation to explanatory and empirical claims. §2 develops the physical interpretation of (S, φ) as a finite lossless memory with partial read-write access, and addresses the substrate objection (what is the memory made of?) by identifying it as gauge in the precise technical sense established by the substratum gauge group of §4. §3 states and proves the reconstruction theorem (Theorem 23). §4 states and proves the substratum gauge group result (Theorem 24) and develops the three-level gauge hierarchy. §5 makes the synthesis claim explicit, citing [Main], [SM], and [GR] for the empirical content. §6 discusses structural realism, the ontological hierarchy, and the dissolution of the measurement problem. §7 concludes. The paper is short by design: the technical content fits in two sections (§§3–4) and the rest is the structural and interpretive context that shows why those two sections matter.

1.1 The metaphysical commitment

The framework's primary metaphysical claim is *ontic structural realism applied to the reconstructed local-sector equivalence class*. What is real at this level is the gauge-invariant local lattice/gauge structure fixed by Theorem 23, not any specific substratum representative. The substratum gauge group (§4) identifies representative freedoms that are observably irrelevant. H-Bell is a separate compatibility condition for embedding the Bell-violating composite structure into the same deterministic ontology; neither Theorem 23 nor Lemma 24.1 proves that the Bell-inclusive completion is unique. A stronger claim — realism about a particular full Bell-complete bijection — therefore outruns the reconstruction theorem.

Stated at its most distilled, the framework's foundational commitment — what [Main §1.2] gives as "observation occurs" — is itself structured: it is two axioms. The first, that tokened differentiation occurs, is indubitable and conceptually primitive. The second, that differentiation recurs — across more than one domain, minimally configuration and time — is a substantive posit: not indubitable, not derivable from the first (a no-go result establishes this), though presupposing it. The commitment is single in the sense that the two axioms invoke one principle; it is, in count, two, and the second is the framework's one non-evidential foundational posit. Part I of the companion paper *Physics Modulo Gauge* itemizes what the first axiom forces as lemmas and what the second adds. (One item of that itemization is a reading of the primitive, adopted here explicitly: the observer's embeddedness — the proper-part decomposition $V \subsetneq S$ — belongs to the first axiom, because "tokened differentiation" is read perspectivally, as a registering from a locus against a remainder. On a thin reading it would instead be a third axiom; the framework adopts the perspectival reading, so the count is two. The choice is presentational, not empirical, and changes nothing downstream of (S, φ, V).)

The structural-realism commitment operates at multiple levels. Within OI's specific universality class, the framework's theorems license realism about $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$. Across structural classes — at the level of partial-trace observational features that any embedded-observer system must respect — broader structural realism applies, with the framework's universality-class equivalence ([Structure §9]) being the appropriate gauge equivalence. The full multi-level structure is articulated in [Structure], which canonicalizes both the observation hierarchy and the gauge hierarchy and their orthogonality, and develops the cross-framework analysis through universality classes of embedded observers. The substratum-class commitment of this paper is one level of a broader hierarchical structural realism: realism at multiple combinations of observation-hierarchy depth and gauge-hierarchy breadth, with the framework's empirical content concentrated at the deepest observation level (Level D in [Structure §2]) and the most class-specific gauge level (Level G3 in [Structure §2]). The articulation here focuses on this concentrated level because that is where the reconstruction theorem and substratum gauge group operate; the

broader hierarchical content is developed in [Structure].

Two positions often proposed as alternatives to structural realism are, in this framework, consequences of it. *Explanatory economy*: if the equivalence class is real, the emergence of quantum mechanics in [Main §3] is a theorem about representations of a real structure, not a coincidence requiring additional interpretation, and the framework's reduction of postulates follows from that uniqueness — under Theorem 23's stated conditions — rather than standing as an independent virtue. *Empirical predictions*: if the class's structure is real, its structural features — the dimension $d = 3$, the $K = 2d = 6$ internal components, the cubic decomposition $(3, 2, 1)$, the wave equation, the cosmological-horizon partition — imply specific numbers at the observable level, giving the twenty-two quantitative predictions of [SM §7] and the gravitational predictions of [GR §7]. The three framings (realist, explanatory, empirical) answer three different questions — what is real, what is gained over standard QM, what is testable beyond QM — rather than providing competing answers to the same question. The rest of this paper develops the structural and explanatory consequences; [SM] and [GR] develop the empirical consequences.

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2. The Physical Interpretation of (S, φ)

Storage and memory. S is the set of all distinguishable states: *finite capacity*. φ is a bijection: *perfect memory* — information is never created or destroyed. Together, (S, φ) is a finite lossless memory. The partition V defines the observer's *access* — both read and write. The observer reads the visible sector and, through the coupling H_{int}, visible operations write correlations into the hidden sector. The slow-bath form of (C2) is one mechanism that preserves those writes until readback. C4 then makes the visible law genuinely history-sensitive; on a fixed finite recurrent representative any non-permutation rooted map, which that readback supplies, forces indivisibility somewhere in the recurrence cycle. The resulting finite observable law has an exact fixed-basis unitary/Born representation by the foundational equivalence theorem. The unresolved step is not the existence or quadratic form of that representation, but its simultaneous extension to the full coherent local instrument/composite algebra. Interference in the mechanism picture is write-then-read: information deposited in the hidden sector during one transition is retrieved later and changes the visible transition statistics.

The 10^{122} CC discrepancy is the compression ratio between total storage and readable storage. The dark sector is the gravitational effect of the unreadable storage. The Bekenstein-Hawking entropy is the storage capacity of the partition boundary. The action scale $\hbar$ is the conversion factor between storage geometry and read-write statistics.

Remark (the domain of the local lattice dynamics). The wave equation of [SM §4.1] — the unique second-order reversible nearest-neighbor dynamics

compatible with center independence, isotropy, and linearity — governs the full **local lattice sector** used by the SM/GR construction, not the visible degrees of freedom alone. Visible and hidden lattice degrees of freedom in that sector run the same wave equation; the partition V imposes observer access without changing the local update. This is enough for (i) the trace-out of [Main] over lattice-hidden degrees of freedom, (ii) the nested local trace-out of [GR §8.4], and (iii) the link-carrier construction of [SM §4.7.1.2]. It is not enough for the Bell-violating deterministic completion under measurement independence: a nearest-neighbor update, whose cone advances one edge per update, together with interventions each reading and writing only its own wing and readouts outside the remote setting's graph cone, falls under [Main §3.3]'s Bell ceiling and gives $|S_{\mathrm{CHSH}}| \leq 2$. The one-object synthesis therefore additionally requires H-Bell (§3.3): the state-dependent graph must realize Bell-relevant preparation-indexed parameter dependence with operational no-signaling while preserving the metric/Ollivier–Ricci continuum limit required by the GR projection.

The substrate objection. "What is the memory made of?" (S, φ) is a *complete description* of reality — it determines all observables. Space, time, matter, and energy are derived from (S, φ), so they cannot appear in its definition without circularity. Whether (S, φ) *is* reality or *describes* reality is provably undecidable by any measurement an embedded observer can perform — an instance of the factoring argument ([GR §6.3]): the two readings determine identical accessible data, and any adjudication procedure available to an embedded observer is a function of that data, so no measurement-based criterion can separate them (the reconstruction theorem below supplies the identical-data premise). The question may have substantive content in mathematical foundations or other access modes not bound by embedded physical observation, but the framework does not address that.

Relation to computation. A Turing machine has a tape (storage), a head (partial read/write access), and a transition function (update rule). The correspondence is suggestive: S is the tape, V is the head's access window, φ is the transition function. But three differences are physically significant. First, a Turing machine's tape is potentially infinite; S is finite — finiteness is essential for the recurrence proof of P-indivisibility, which holds on every finite reversible representative with a non-deterministic visible step; the representation itself is universal, though the effective finiteness result ([GR, §8.4]) shows the deep hidden sector may be infinite without affecting the leading-order transition data on accessible timescales. Second, a Turing machine is generally irreversible (it can erase, overwrite, and halt); φ is a bijection — nothing is erased, nothing is created, there is no halting. Third, a Turing machine computes an *extrinsic function* (input $\rightarrow$ output for an external user); (S, φ) computes no extrinsic function — it is a closed permutation that cycles through states and returns. The appearance of dynamics, probability, particles, and forces is entirely the observer's perspective — what the permutation looks like through the partial window V.

Remark (the Turing connection under effective finiteness). The three differences soften under the effective finiteness result. With the deep hidden sector potentially infinite, the first difference is between the formal definition (S finite) and the physical requirement for accessible-timescale content (only $\mathcal{C}_V \times \mathcal{C}_B$ need be finite). The second difference is a specialization, not an opposition: reversible Turing machines (Bennett, Fredkin-Toffoli) are a well-studied subclass. The third difference — extrinsic vs. intrinsic — is the one that does physical work. The mapping is then structural: V is the head, H is the tape, φ is a reversible transition function, and C1–C4 characterize the architecture. The framework extends Turing's question: instead of asking what a machine can compute for an external observer, it asks what computation looks like to a component of the machine — and proves the answer admits the quantum representation universally — the memory-bearing sector, where the distinctly quantum phenomenology lives, characterized by C1/C3/C4.

The arrow of time. The substratum has no arrow of time — φ and φ^{-1} are equally valid. Observers satisfying C1–C4 are nonetheless structurally restricted to the non-equilibrium phase of (S, φ): the theorem in [Main, §4.6] shows that $\tau_B(V)$ is bounded by the local mixing time on equilibrium microstates, so C2 fails there — and hence C4, by the erasure lemma of [Main §2.3], which is the condition the characterization registers. This replaces the low-entropy "past hypothesis" with a structural observer-selection rule and excludes Boltzmann brains structurally rather than anthropically. Entropy increase is then emergent as a coarse-grained description of the non-equilibrium dynamics observers necessarily see.

The incompleteness family. The framework's central result belongs to a family of impossibility-with-structure theorems. Gödel: a formal system cannot prove all truths about itself — the unprovable truths have rigid structure. Turing: a computer cannot decide all questions about its own behavior — the undecidable problems have rigid structure. OI: an observer embedded in a deterministic system cannot access the complete state — the emergent description has rigid structure (quantum mechanics). The common structure is *self-reference under finite resources*.

The precise OI analog of the halting problem is: *can the observer determine the hidden-sector state h?* The question is well-posed — h has a definite value at every moment, because (S, φ) is deterministic. Different h values produce different physical outcomes. But the observer provably cannot determine h: multiple hidden states are compatible with any visible-sector history, and transition probabilities $T_{ij}(t)$ are averages over h. The structural consequence of this inaccessibility is quantum mechanics — just as the structural consequence of the halting problem's undecidability is computability theory. The framework also identifies a second class of inaccessible quantities — the alphabet size q ([SM, §2.7]) and the deep-sector cardinality $|\mathcal{C}_D|$ ([GR, §3.2]) — but these are *gauge*, not undecidable: different cardinalities of a deep sector that no dynamics couples to the boundary layer produce identical observables (Theorem 24(iii),

§4), and so, under the alphabet-size hypothesis of [SM, §2.7], do different values of q (Theorem 24(ii)), so the question is physically empty. For a deep sector coupled to the boundary layer with slow dynamics, different values of $|\mathcal{C}_D|$ produce the same leading-order transition law on accessible timescales (Corollary (effective finiteness and gauge-class transfer), §4), and the question is empty only within that law. The hidden state h is undecidable (real answer, provably inaccessible). In the decoupled case the cardinality $|\mathcal{C}_D|$ is gauge (no answer to find).

Mathematics and physics. The trace-out performs a Jordan-Chevalley projection ([SM, Appendix A]): it extracts the semisimple part of the dynamics and erases the nilpotent monodromy. Physics is the semisimple shadow of mathematics — the diagonalizable spectral data, projected by the trace-out and organized by the gauge group's representation structure. The reconstruction theorem (below) proves — under its stated conditions — that the **local lattice/gauge residue** is determined up to gauge. It does not prove uniqueness of the Bell-inclusive full substratum.

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3. The Reconstruction Theorem

The forward direction — from (S, φ) to observed physics — is established by [SM §§3.1, 4–5] and the companion paper [Main]. The converse question is which parts of (S, φ) the observed physics fixes. The reconstruction proceeds in three stages. The composite theorem (Theorem 23) aggregates the stages and separates the uniquely reconstructed local lattice/gauge residue from the Bell-inclusive completion, whose existence is conditional on H-Bell and whose uniqueness is open.

3.1 Inputs to the reconstruction

The reconstruction takes two kinds of inputs: empirical observations and structural assumptions. Being explicit about both is essential to the uniqueness claim.

Empirical inputs (facts about the observed universe):

(E1) **Unitary quantum mechanics.** The observed physics is described, to the precision tested, by quantum mechanics: states as vectors in a complex Hilbert space, unitary time evolution, self-adjoint observables, Born-rule outcome statistics.

(E2) **Bell violations.** The observed correlations violate Bell inequalities, ruling out local hidden-variable theories with factorizable distributions.

(E3) **Finite boundary entropy.** The entropy of any bounded region is finite and scales as the area of its boundary. This is a theoretical input — the

holographic bounds [1, 2], which are not themselves observed — applied to the cosmological horizon, whose finite area makes the bound apply.

(E4) **Spatial isotropy.** Observed physics is rotationally invariant; no spatial direction is preferred.

(E5) **Propagating gravity.** Gravitational degrees of freedom propagate (gravitational waves are observed). This requires $d \geq 3$, since the Weyl tensor vanishes for $d \leq 2$. (Used in Stage 2(a) below; one of the three independent forward filters that select $d = 3$.)

(E6) **Stable matter.** Atoms, planets, and bound states exist on observed timescales. This requires $d \leq 3$, since the Coulomb potential in d dimensions gives unstable atoms for $d \geq 4$ (Ehrenfest [10]) and gravitational orbits are unstable for $d \geq 4$ (Tangherlini [11]). Without stable matter, no embedded observers exist to define the partition. (Used in Stage 2(a); second forward filter for $d = 3$.)

(E7) **Boundary-entropy concordance.** The cosmological parameter inferred from observation under the standard cosmological model satisfies $\rho_s/\rho_{\text{crit}} \approx 1$ (spatial flatness near critical density); it is an inferred quantity, not a directly measured one. In d spatial dimensions, the general-d form of this ratio is $\rho_s/\rho_{\text{crit}} = 2/(d-1)$, which equals unity only for $d = 3$. (Used in Stage 2(a); third forward filter for $d = 3$. See [SM §3.2] and [GR §7.2] for the derivation.)

Structural assumptions (restrictions on the class of candidate substrates (S, φ)):

(A1) **Finiteness.** The configuration space S is finite. (Two-part status. E3, with the holographic bound read as a Hilbert-space dimension cutoff $\dim \mathcal{H} \leq e^{A/4}$, bounds what observation reaches: the visible sector and its boundary layer, $\mathcal{C}_V \times \mathcal{C}_B$ — on which the emergent observables depend at leading order on accessible timescales, by the boundary-only dependence lemma [GR §3.2]. For a deep hidden sector that no dynamics couples to the boundary layer, the extension to all of S is a gauge choice rather than a further fact: by the deep-sector enlargement freedom (Theorem 24(iii)), such a sector may be finite of any size, or infinite, without changing any transition probability, and A1 selects the minimal — finite — representative of the equivalence class. A deep sector coupled to the boundary layer with slow dynamics changes the transition probabilities at $\mathcal{O}(t/\tau_B)$; its size is not resolved by the leading-order transition law on accessible timescales, which is the scope of the transfer in the corollary following Theorem 24. In the decoupled case the substrate's total cardinality beyond the boundary layer is not an observable question, in the same sense that the uniform vacuum offset is not.)

(A2) **Determinism.** $\varphi : S \to S$ is a bijection (deterministic, reversible dynamics). (Two-part status, per the dilemma argument of [Main §2.2]: a non-injective descent either leaves a statistical trace — excluded where the observed statistics are of the doubly stochastic class that the uniform-prior marginal of a bijection

produces — or leaves none, in which case it is removable, every map on a finite set restricting to a bijection on its eventual image, where all recurrent registered history lives. Bijectivity for the dynamics governing registered history is thus partly structural and partly anchored by one empirical input, the observed statistics in that scope; Main's honest-accounting ledger records the same division.) The anchor is empirically live: its current frontier is nanoparticle matter-wave interference at macroscopicity $\mu = 15.5$ — sodium clusters above 1.7×10^5 amu held in 133-nm path superpositions [4] — an order of magnitude beyond the prior benchmark, and the direct one-sided test against the intrinsic-collapse alternatives that would break it.

(A3) **Bounded coupling degree.** Each site is coupled to a bounded number of neighbors in φ's action. (Required for locality and the emergence of a coupling graph with well-defined dimension. The physical content is boundedness itself; the specific degree is gauge — any bounded-degree graph in the quasi-isometry class produces the same emergent quantum-mechanical, dimensional, and gravitational content, Theorem 24(iv).)

(A4) **Center independence.** The dynamics φ does not depend on a choice of "center" site, in two senses used below: φ commutes with lattice translations up to gauge, and the update at a site contains no explicit copy of that site's present value ([SM §4.1]). (The second sense is what the wave-equation derivation of Stage 2 uses. Its observational anchor is the homogeneity of physical law — no experiment has found a preferred location in the laws — the cosmological principle's complement to E4's isotropy; whether it is independently warranted by the observer architecture remains the question flagged in §1.)

(A5) **Linearity.** The wave equation for φ is linear. (Equivalent to amplitude-scale gauge invariance — that the field-value scale is unphysical — via the linearity-equivalence lemma of [SM §4.1]; hence necessary *given* that gauge principle and "partly derived" in exactly that sense. Whether the q-size gauge freedom of [SM §2.7] entails amplitude-scale gauge is an identified open step; pending it, nonlinear alternatives are excluded precisely to the extent amplitude-scale gauge is assumed, and would otherwise require a separate derivation.)

(A6) **Background independence.** The dynamics is covariant under spatially-varying internal-index transformations: a site-dependent transformation $G(\mathbf{n})$ of the internal index is a symmetry of the dynamics provided the coupling data carried on the links are transformed with it, $M(\mathbf{n}, \hat{e}_j) \to G(\mathbf{n})\, M(\mathbf{n}, \hat{e}_j)\, G(\mathbf{n} + \hat{e}_j)^{-1}$ ([SM §3.1]). The content of the assumption is this covariant interface — the coupling is link-valued data on which the transformations act — and once it is in place the local transformation law imposes no further condition on the rule. A6 is not the stronger fixed-background condition that the same transformations leave the dynamics invariant with the coupling held fixed and preserved pointwise: for an invertible constant coupling that condition forces the transformation to agree across every coupled pair of sites, and at the symmetric point of the coupling matrix it fails for every transformation that differs across a

coupled pair. The global commutant symmetry — the transformations constant across the lattice that leave the coupling itself unchanged, the specialization of A6 in which nothing needs transporting — is what the local gauge reading of [SM §3.1] proceeds from. The state-dependent coupling graph of [SM §3.1], under which the graph itself evolves with the state, is a separate principle that shares the name and is not A6.

The theorem's uniqueness claim holds under E1–E7 **and** A1–A6 jointly; removing any of A3–A6 either requires new derivations or weakens the uniqueness. The set A1–A6 is sufficient for the reconstruction but is not proved to be minimal: A4 (center independence) and A6 (background independence) overlap in physical content (A6 promotes the symmetry that A4 constrains), and A5 (linearity) is equivalent to amplitude-scale gauge invariance via the linearity-equivalence lemma of [SM §4.1], hence partly derived — necessary *given* amplitude-scale gauge, with whether [SM §2.7]'s q-size gauge entails it left as an identified open step. A tighter axiomatization may be possible but is not pursued here; the reconstruction's validity depends on the sufficiency of A1–A6 and E1–E7, not on their independence. The empirical inputs E5, E6, E7 (propagating gravity, stable matter, boundary-entropy concordance) provide explicit forward filters for $d = 3$ — see Stage 2(a) below.

Critical dependencies. Theorem 23's proof relies on the following prior theorems, whose individual correctness is assumed:

- [Main §2.3] history readback for the **memory-bearing temporal** sector used in Stage 1 (a modeling premise about the temporal data, not an inference from Bell violation)
- [Main §3.3] Bell ceiling and the adopted measurement-independent, ontically parameter-dependent Bell branch (a separate spatial-composite premise; Bell violation is not used as a witness of temporal P-indivisibility)
- [Main §3.2] Forward Stinespring dilation (bijection → CPTP channel) and reverse-direction lemma (stochastic process → bijection)
- [Main §3.4] Characterization theorem (accessible non-Markovianity — the framework's finite-horizon visible-law memory/backflow notion, not generic quantum non-Markovianity — $\iff$ C1/C3/C4 per finite accessible horizon; C2's necessity conditional on the hidden-sector mixing hypothesis, ETH its physical motivation)
- [SM §3.2] Coupling-graph dimension → d = 3
- [SM §4.1] Center independence + isotropy + linearity → wave equation
- [SM Theorems 5–15] Gauge group, generations, hypercharges derivation chain
- [SM Theorems 17–21] T-invariance → $\theta \in \{0, \pi\}$ and reciprocal transition counts; $\bar{\theta}$ open (H-top, H-det)
- [GR §3] Gap equation from thermal self-consistency → $\hbar = c^3\epsilon^2/(4G\sigma)$

Theorem 23's confidence is bounded above by the minimum confidence in these dependencies.

3.2 Stage 1: Observed QM → deterministic embedding with C1–C4

Inputs: E1 (unitary QM), E2 (Bell violations), E3 (finite boundary entropy), A1 (finiteness), A2 (determinism), and M1, in two clauses: **M1-T**, the observed temporal quantum data used in Stage 1 lie in the accessible history-readback sector of [Main §2.3–§3.4]; and **M1-B**, Bell-violating composites take [Main §3.3]'s measurement-independent, ontically parameter-dependent branch with operational no-signaling. M1-T and M1-B are logically distinct; Bell violation is not taken to imply temporal P-indivisibility.

Output: There exists a triple (S, φ, V) with S finite, φ a bijection on S, and $V \subset S$ a distinguished subset such that: - The observed quantum dynamics is the reduced description obtained by tracing out $S \setminus V$ from the deterministic evolution under φ. - The triple satisfies C1 (non-trivial coupling), C2 (record persistence — the slow bath one realization), C3 (sufficient hidden memory capacity), and C4 (history readback) for the M1-T temporal sector: [Main §3.4]'s universal hidden-memory theorem — accessible non-Markovianity $\Longleftrightarrow$ C1/C3/C4 realization per accessible horizon, with $M_t > 0$ forcing unavoidable hidden predictive memory — makes readback-relevant hidden memory unavoidable in every faithful deterministic realization. M1-B separately requires Bell-relevant cross-wing coupling/parameter dependence and does not supply C4.

Conditional structure. C1, C3, and C4 for the memory-bearing temporal sector are derived from E1–E3, A1–A2 together with **M1-T** — C4 via the hidden-memory theorem applied to the stipulated accessible temporal memory/P-indivisibility; conditional on M1-T, they hold without further hypothesis. **M1-B** is separate: it maps E2 to the ontically parameter-dependent Bell composite and supplies an independent cross-wing-coupling requirement, not C4. C2 is conditional on the quantitative mixing hypothesis of [Main §3.4] — uniform total-variation closeness of the hidden sector's post-relaxation conditionals to a single bath distribution, under which the divisibility deficit obeys an explicit bound $(k-1)\varepsilon$; without it, the necessity direction (P-indivisibility on accessible timescales forces $\tau_S \ll \tau_B$) cannot be established. ETH is the physical case for the hypothesis, as discussed in [Main §3.4]'s "Remark (Status of ETH)." ETH is a well-supported conjecture for generic chaotic many-body Hamiltonians and is the standard assumption for the cosmological-horizon complement; relaxations or failures of ETH would require a separate argument. The framework's gauge-group derivation through Stage 2 inherits this conditional.

Bell–gravity compatibility hypothesis (H-Bell). The Stage-2 reference nearest-neighbor wave-equation graph has a pointwise causal cone advancing one edge per update; whether that cone or the lift's long-wavelength speed $1/\sqrt{3}$ is the physical c is not derived [Main §3.3]. Under measurement independence, setting interventions that each read and write only their own wing and ± 1 readouts outside the remote setting's graph cone, [Main §3.3] therefore makes that **reference graph** Bell-local and caps CHSH at 2. The framework neverthe-

less already makes the graph state-dependent: the discrete Einstein equation of [SM §3.1] is formulated on $G(x)$, and its mod-q spatial-Markov proof explicitly assumes uniform initial conditions. M1-B may therefore be implemented by **preparation-indexed adjacency**: the vacuum/uniform reference class retains the geometric bounded-range graph used by the local SM/GR derivation, while an entangling preparation supplies bounded-degree Bell-relevant long edges (or an equivalent ontically nonlocal response map) furnishing parameter dependence without operational signaling.

H-Bell is the remaining compatibility statement for that branch: the preparation-indexed graphs must (i) supply the required ontic parameter dependence, (ii) preserve operational no-signaling, and (iii) satisfy a metric/Ollivier–Ricci stability condition strong enough that the continuum-curvature step of [SM §3.1] remains valid on the state classes to which the Einstein reconstruction is applied. The graph-boundary area law is not the primary unresolved issue: $|\partial V|$ is defined on $G(x)$ and therefore counts preparation-induced crossing edges. The open step is the identification of that graph boundary/curvature with continuum geometric area/Ricci curvature. [SM §3.1]'s sparse local-contamination estimate, $L_N m_N/N \to 0$, is a sufficient condition for typical mesoscopic neighborhoods to remain reference-like, but it does not prove global curvature convergence, control defect endpoints, or establish the integrated-curvature limit. H-Bell is therefore an explicit curvature-stability/composite-existence hypothesis, not a consequence of A1–A6 or of the nearest-neighbor wave equation.

Two points limit the exposure this conditional creates. First, the relevant question is not whether ETH holds universally — it is known to fail for integrable and many-body-localized (MBL) systems — but whether it holds for *the* hidden sector implied by the framework's other commitments. The hidden sector here is the high-capacity complement ($N_H \gg N_V$ — the comfortable realization of C3's capacity floor) of a bounded-coupling bijection on a statistically isotropic graph (A3, E_isotropy). Integrability and MBL are both non-generic in a specific sense relevant here: integrability requires an extensive set of local conserved quantities, which a statistically isotropic bounded-degree coupling graph without fine-tuned commuting structure does not possess; MBL requires strong quenched disorder sustaining localization against the thermalizing tendency of generic interactions, and is moreover known to be unstable in dimension $d \geq 2$ (the avalanche instability). Since the reconstruction independently selects $d = 3$ (§3.3, [SM §3.2]), the hidden sector sits in the regime where MBL does not survive and generic interacting dynamics thermalize — so ETH is the expected case, not an additional fine-tuning. Second, the conditional attaches only to the *necessity* direction of C2; the *sufficiency* direction — C1–C4 realization yields the memory-bearing non-Markovian phenomenology, whose finite law universally admits the quantum representation — is unconditional, so the framework's positive predictive content does not rest on ETH. ETH is required for the necessity direction — arguing that the observed memory-bearing phenomenology *forces* persistent hidden structure, realized as the slow bath, not to

argue that the slow-bath structure produces the quantum phenomenology. A referee skeptical of ETH may therefore read the reconstruction's uniqueness as holding within the class of ETH-satisfying (equivalently, non-integrable, non-MBL) hidden sectors — a class the framework argues is generic at $d = 3$ rather than special.

Derivation. The observed quantum dynamics provides, for each finite experimental context, a stochastic law on a finite visible configuration space. By [Main]'s exact finite-horizon equivalence, this law has a finite reversible deterministic realization and an exact fixed-basis unitary/Born representation. The representation step is universal and therefore does not itself reconstruct the nontrivial OI sector. The additional modeling premise M1 separates two empirical structures. **M1-T** places the temporal data used for Stage 1 in the accessible history-readback sector; [Main §2.3] then proves that on a fixed finite representative a non-permutation rooted map, which C4 readback supplies, forces indivisibility somewhere in the recurrence cycle. **M1-B** places the Bell data on [Main §3.3]'s measurement-independent, ontically parameter-dependent branch; Bell violation therefore certifies Bell-relevant cross-wing coupling but is not used to infer temporal P-indivisibility. By [Main]'s finite-horizon equivalence ($S \Rightarrow D$, the process-dilation and response-table constructions of [Main §3.4]), the temporal law admits a realization as the visible marginal of a deterministic bijection φ on an enlarged finite space $\mathcal{C}_V \times \mathcal{C}_H$, with a hidden prior independent of the visible initial state carried as part of the realization datum; for a single one-step doubly stochastic matrix the reverse-direction lemma ([Main §3.2]) gives the uniform-prior form, exactly when its entries are rational and to arbitrary precision otherwise. This delivers the triple (S, φ, V) with $S = \mathcal{C}_V \times \mathcal{C}_H$ and $V = \mathcal{C}_V$. Non-permutation $T(t)$ in the M1-T temporal sector gives the corresponding C1 witness; independently, M1-B's ontic parameter dependence requires Bell-relevant cross-wing coupling. The characterization theorem [Main §3.4] establishes that the resulting dynamics, conditional on the hidden-sector mixing hypothesis ([Main §3.4]; ETH is its physical motivation), necessarily satisfies C2 (record persistence is required for the non-Markovian returns that produce quantum interference) and C3 (sufficient hidden-sector capacity is required for the data-processing bound to accommodate observed information backflow).

Stage-1 gauge statement. For the **finite channel/dilation datum actually used by the temporal reconstruction**, the familiar dilation freedoms are absorbed into the stated substratum-gauge directions: hidden-sector basis/relabeling freedom and observationally inert deep-sector enlargement. This is not uniqueness of the full triple (S, φ, V) across all coherent composites. In particular, it does not select a preparation-indexed Bell adjacency rule, and the operational lift still requires its separately stated finite-predictive-dimension, local-tomography and purification hypotheses.

Status: Theorem for the scoped finite dilation datum; no Bell-inclusive Stage-1 uniqueness claim.

3.3 Stage 2: Embedding + isotropy → specific lattice + gauge structure

Inputs: Stage 1 output + E4 (spatial isotropy) + E5 (propagating gravity) + E6 (stable matter) + E7 (boundary-entropy concordance) + A3 (bounded coupling) + A4 (center independence) + A5 (linearity) + A6 (background independence) + max-lattice-speed condition (for the wave-equation coupling constant; see step (b) below). Identification of this local lattice output with the Bell-violating Stage-1 completion additionally assumes H-Bell.

Output: The local propagating substratum sector is a $d = 3$ cubic lattice bijection with: - $K = 2d = 6$ internal components per site - Coupling matrix eigenvalue multiplicities $(3, 2, 1)$ - Gauge group SU(3) × SU(2) × U(1) - Three generations of chiral fermions in the SM representations - One Higgs doublet $(\mathbf{1}, \mathbf{2}, +1/2)$ - Anomaly-free hypercharges $(Y_Q, Y_u, Y_d, Y_L, Y_e) = (1/6, 2/3, -1/3, -1/2, -1)$

Derivation.

(a) *Dimension.* The coupling graph inherited from Stage 1 has polynomial growth exponent d. Three independent forward filters from the empirical inputs converge on $d = 3$: propagating gravity (E5) requires $d \geq 3$ (the Weyl tensor vanishes for $d \leq 2$); stable matter (E6) requires $d \leq 3$ (Coulomb instability of atoms for $d \geq 4$, plus Tangherlini's gravitational orbit instability); boundary-entropy concordance (E7) gives $\rho_s/\rho_{\text{crit}} = 2/(d-1) = 1$ at $d = 3$ uniquely. The intersection of the first two filters already pins $d = 3$; the third is a sharpening consistency check that the framework passes exactly. The integer-polynomial growth assumption (rather than fractal or exponential growth) is itself derived from self-consistency: exponential growth fails Lorentz-invariant dispersion (Jacobson's GR derivation requires it); fractal growth fails the cubic-lattice symmetry needed for the cubic-group decomposition in step (d) below. By Gromov's theorem and statistical isotropy (E4), the surviving graph class is quasi-isometric to $\mathbb{Z}^d$ for integer d; the three filters then select $d = 3$. ([SM §3.2] presents this argument in full; the inputs E5, E6, E7 surfaced here make explicit the empirical content the filters require.)

(b) *Wave equation.* Center independence (A4, in its functional sense: no self-term), isotropy (E4), and linearity (A5) uniquely select, within the nearest-neighbor class (an assumption beyond A3's bounded degree; see the scope remark at [SM §4.1]), the wave equation up to a multiplicative coupling constant α ([SM §4.1, Theorem]): the unique second-order linear dynamics on a lattice that is translation-invariant, isotropic, reversible and free of a self-term has the form $f = \alpha(x_1 + x_2 + \cdots + x_{2d}) \bmod q$ with coupling coefficient α. The coefficient is neither the propagation speed nor free to be set to 1: causal support travels at most one lattice edge per update whatever its value, and on the observer-level normalized branch it equals $1/d$ ([SM §4.1, Remark]).

(c) *Internal components.* Under H-link the link count gives $K = 2dm$, with $m = 1$ giving $K = 6$; coupling-degree minimization does not establish it (see 6). This locks the candidate-sector count at three — three physical generations only under H-spin' ([SM §4.7]) — (three spin-1/2 staggered tastes emerge from the $K = 6$ minimum).

(d) *Gauge group.* Cubic-group decomposition of the $K = 6$ components gives multiplicities $(3, 2, 1)$ (Theorem 7). The unitary group of the bicommutant of the cubic-group action on $V_K = V_3 \oplus V_2 \oplus V_1$ — equivalently, the stabilizer of any generic O-equivariant operator on V_K — is $U(3)\times U(2)\times U(1)$. (The strict pointwise commutant of the action is $U(1)^3$ by Schur's lemma; the gauge group is the unitary group of the block algebra $\mathrm{Mat}(3) \oplus \mathrm{Mat}(2) \oplus \mathrm{Mat}(1)$ generated by the action.) The reduction to $SU(3) \times SU(2) \times U(1)$ occurs because the overall $U(1)$ phase of each block is gauge under *amplitude-scale invariance* (Theorem 24, Remark below): a global rescaling of the V_3 block by a phase is indistinguishable from a global rescaling of the field value ϕ, which amplitude-scale gauge declares unphysical. (This is the field-value-scale freedom — the additive-automorphism relabellings of the alphabet — *distinct* from the alphabet-*size* freedom of generator (ii).) The same applies to the V_2 block. The singlet-block $U(1)$ survives this stripping (it cannot be absorbed into alphabet-freedom because it acts non-trivially on the other blocks via the determinant constraint) and becomes the hypercharge $U(1)$. Background independence (A6) — covariance under site-dependent transformations, the link coupling transformed with them — then promotes the remaining global $SU(3) \times SU(2) \times U(1)$ to local gauge invariance ([SM §3.1]).

(e) *Matter content.* Staggered reduction gives three degenerate spin-1/2 sectors (Theorems 8–10). Fermion embedding is completed via the link-carrier construction ([SM §4.7.1.2]). Partition-spinor identification (Theorem 12) and trace-out (Theorem 13) give chirality. Anomaly cancellation (Theorems 14–15) uniquely fixes the hypercharges.

(f) *Discrete symmetries.* T-invariance of the bijection φ gives $\theta \in \{0, \pi\}$ and reciprocal visible transition counts at every scale (Theorems 17–21); it does not fix $\bar{\theta}$, which remains open under H-top and H-det [SM §5]. Reciprocity leaves the Jarlskog invariant free; only a mechanism making Y_u and Y_d simultaneously real would force $J = 0$, which is the falsification control on H-det proposals [SM, §5.3–5.5].

Uniqueness at Stage 2. Each step (a)–(f) is an *if-and-only-if*: the stated structure is the unique consistent choice given the inputs. The full chain is therefore a composition of unique selections, so the output is unique up to $\mathcal{G}_{\mathrm{sub}}$.

Bell compatibility. Step (b)'s nearest-neighbor **reference** cone cannot itself realize M1-B, while retaining measurement independence, in a Bell protocol whose wings are separated by more than the graph cone over its duration. Stage 2 therefore establishes the local lattice/gauge sector on the geometric reference

class. The state-dependent $G(x)$ machinery leaves a native route for M1-B through preparation-indexed adjacency, but the Ollivier–Ricci continuum theorem presently cited by [SM] does not cover those long-edge perturbations. Nor does it cover the unperturbed reference graph, and there the situation is worse than open: the degree-6 cubic hop metric is exactly ℓ_1, with a scale-independent $\sqrt{2}$ diagonal stretch, so the hop-metric curvature route fails for the reference family and no $D = 3$ extension of the cited theorem would rescue it. The repair is to re-found curvature on the propagation/Laplacian geometry, which the substratum's isotropic leading dispersion already supplies [SM §3.2]. Promoting the local sector and M1-B to one common (S, φ) is therefore conditional on H-Bell's curvature-stability requirement.

Status: Theorem for the local lattice sector; one-object Bell compatibility conditional on H-Bell.

3.4 Stage 3: Dynamics → emergent fundamental constants

Inputs: Stage 2 output.

Output: The emergent action-scale relation $\hbar = c^3\epsilon^2/(4G\sigma)$, equivalently $\epsilon = 2\sqrt{\sigma}\, l_p$, where l_p is the Planck length and σ the entropy per boundary mode ([GR §3.1]); it fixes only $\epsilon/\sqrt{\sigma}$, not the value of $\hbar$ ([GR §4]).

Derivation. The gap equation from thermal self-consistency of the boundary layer ([GR §3]) gives the stated relation. The derivation uses: existence of thermal equilibrium at the boundary layer (a consequence of the C1–C4 dynamics from Stage 1), the boundary-only dependence lemma ([GR §3.2]), and the infrared detailed-balance condition H-slope with the horizon and frame conditions ([GR §2, §3.2, §8.5]), which does not follow from φ being a bijection ([GR §3.2]) and is an assumption beyond Stages 1–2.

Uniqueness at Stage 3. The gap equation admits a unique solution under the stated inputs ([GR §3.2]).

Status: Conditional on H-slope with the horizon and frame conditions ([GR §3.2]).

3.5 The composite theorem

Lemma 23.0 (Local-sector uniqueness preservation). *Let (S_1, φ_1) and (S_2, φ_2) both reproduce the same inputs E1–E7 and satisfy A1–A6 and M1, with the hidden-sector mixing hypothesis where required. Conditional on Lemma 24.1's semigroup-transfer/completeness step, their reconstructed* ***local propagating lattice/gauge sectors*** *have the same Stage-2 invariants up to the local action of $\mathcal{G}_{\mathrm{sub}}$: $d = 3$, $K = 6$, multiplicities $(3, 2, 1)$, the SM gauge group and associated local representation data, and, conditional on H-slope with the horizon and frame conditions of [GR §3.2], the same Stage-3 action-scale relation $\hbar = c^3\epsilon^2/(4G\sigma)$.*

Proof. Stage 1 fixes the finite channel/dilation datum used by the reconstruction

only up to its stated dilation/gauge freedoms and the explicit operational-lift conditions. Stage 2 then selects the same local cubic propagating sector and discrete representation-theoretic invariants from E4–E7 and A3–A6; graph relabeling and deep-sector enlargement lie in $\mathcal{G}_{\mathrm{sub}}$, and alphabet freedom does under the alphabet-size hypothesis of [SM §2.7] (Theorem 24(ii)). Under H-slope, Stage 3 fixes the same gap-equation relation $\hbar = c^3\epsilon^2/(4G\sigma)$; an alphabet change can change the bound $\sigma \leq \ln q$, and the equality of $\hbar$ across it is part of that hypothesis. These statements compare the reconstructed local sector only. They do **not** compare preparation-indexed Bell edge rules or other Bell-nonlocal composite data, because neither Lemma 24.1 nor the Stage-2 uniqueness argument acts on that extra structure. □

Theorem 23 (Layered local reconstruction with Bell compatibility). *Let E1–E7, M1-T, M1-B, A1–A6, the hidden-sector mixing hypothesis for the C2 necessity direction, and Lemma 24.1's semigroup-transfer/completeness step hold. Then the reconstruction uniquely fixes, up to the stated substratum-gauge freedoms, the* ***local propagating lattice/gauge sector*** *with $d = 3$, $K = 6$, coupling-matrix multiplicities $(3, 2, 1)$, gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, the stated local matter/representation content and discrete-symmetry outputs, and $\hbar = c^3\epsilon^2/(4G\sigma)$ conditional on H-slope with the horizon and frame conditions of [GR §3.2], with the same qualifications stated in [SM] for H-spin', H-χ', and the CKM-CP fork. A single measurement-independent deterministic completion that also realizes the Bell-violating M1-B composites exists only conditional on H-Bell. The theorem does* ***not*** *establish uniqueness of that Bell-inclusive full completion.*

Proof. Stage 1 supplies the finite reversible/dilation data and the M1-T temporal conditions used by the local reconstruction, subject to its explicit operational-lift hypotheses and the C2-mixing condition. Stage 2 uniquely fixes the listed local lattice/gauge invariants from those data and E4–E7; Stage 3, conditional on H-slope, fixes the relation $\hbar = c^3\epsilon^2/(4G\sigma)$. Lemma 23.0 preserves that uniqueness at the local-sector level. M1-B adds a separate Bell-relevant requirement. H-Bell states the compatibility/existence condition for realizing it through the state-dependent graph or an equivalent ontically nonlocal response implementation while retaining the continuum-curvature limit. No theorem in the present chain selects a unique preparation-indexed Bell edge rule, and Lemma 24.1 controls the visible channel-semigroup dilation datum rather than such Bell-composite structure. Therefore Bell-inclusive existence is conditional and Bell-inclusive uniqueness remains open. □

3.6 Remarks

Remark (Evidential weight of the reconstruction). The reconstruction uniquely derives $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ with multiplicities $(3, 2, 1)$, three generations (physical reading conditional on H-spin', [SM §4.7]; chirality on H-χ', [SM §4.8]), one Higgs doublet $(1, 2, +1/2)$, and unique hypercharges. Because the derivation is tightly constrained — no fitted parameters, no model-building choices, no

landscape of alternatives at the level of the gauge sector — every independent experimental confirmation of the Standard Model's gauge structure is retroactive evidence for the derivation chain that produces it, with the strength of that evidence bounded by the chain's open links (the construction delivers a distinguished low-freedom representation rather than a uniquely selected one, and the family pattern's selection is not supplied; [SM] Theorem 15 Remark). The Standard Model is the most precisely tested mathematical structure in the history of science ($g-2$ of the electron confirmed to 10^{-12}, electroweak precision observables to 10^{-5}, QCD cross-sections across decades of energy). This experimental record confirms the Standard Model structure; it transfers to the framework only to the extent that the derivation is unique, since the transitivity "data confirms structure, structure is uniquely derived, therefore data confirms the derivation" is only as strong as its middle premise. That premise is conditional on the argued — not proven — sufficiency of the A1–A6 class (Remark on hypothesis dependencies below) and is established operationally by the classified retrodiction record of [SM §7] (two parameter-free structural and five conditional structural entries among twenty-two classified observables) (Remark on finite observation sets below), not by the Standard Model precision record itself. The precision record is therefore necessary but not sufficient evidence for the *uniqueness* of the derivation — which is the framework's distinct claim, and the proper locus of skeptical scrutiny.

Remark (Substantive content of the uniqueness claim). A reasonable objection to any reconstruction theorem whose answer is an equivalence class under a large gauge group is that the result risks tautology: if $[(S,\varphi)]/\mathcal{G}_{\text{sub}}$ is *defined* as the observational-equivalence class, then "observed physics uniquely determines the observational-equivalence class" is close to true by construction, and the gauge group $\mathcal{G}_{\text{sub}}$ here is enormous (state relabeling, alphabet change under the alphabet-size hypothesis, deep-sector enlargement of arbitrary size, graph isomorphism up to statistical class — see §4). The objection has force against a vacuous version of the claim and must be answered by exhibiting the *non-tautological residue*: the specific structure the theorem fixes that does not follow merely from "is observationally equivalent to the data." That residue is a short list of discrete invariants. The theorem does not merely return "whatever is observationally equivalent to observed physics"; it returns specific integers and discrete structures — the spatial dimension $d=3$, the internal-component count $K=6$, the coupling-matrix eigenvalue multiplicities $(3,2,1)$, the gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$, the generation count exactly three, one Higgs doublet, and the anomaly-free hypercharge assignment. None of these is a tautological consequence of observational equivalence: a priori the reconstruction could have returned any dimension, any gauge group, any generation count, or a continuum of them, and the content of the theorem is that the structural assumptions force *these specific discrete values* and no others. The large gauge group $\mathcal{G}_{\text{sub}}$ is quotiented away precisely so that what remains invariant is this discrete list — the continuous and combinatorial freedoms (relabeling, alphabet under the alphabet-size hypothesis, deep-sector size, graph realization) are exactly the di-

rections along which the answer is *not* fixed and should not be, because they are unobservable. The uniqueness claim is therefore substantive in the dimensions that carry physical content (the discrete structural invariants) and silent in the dimensions that do not (the gauge directions): a tautological theorem would fix nothing specific, whereas this one fixes a particular finite list of integers and groups that the framework then matches against the observed Standard Model.

Refinement (which residue is forced, and which is selected). The defense above is correct but must be stated with one further distinction, or it over-claims. Of the discrete invariants listed, those that are *forced by representation theory given the lattice* — the multiplicities $(3, 2, 1)$ and the gauge factors, the generation count, the hypercharges — are non-tautological in the full sense: a priori they could have come out otherwise, and the octahedral character table fixes them. But two of the listed invariants are not forced by the construction in this way: the spatial dimension $d = 3$ is selected by the empirical filters E5–E7 (§3.3), and the simple-cubic Bravais structure is selected because the observed gauge group contains SU(3) (Theorem 7b, [SM §4.6]) — only the cubic crystal system supplies the three-dimensional point-group irrep an SU(3) factor requires. For these, the inference runs from observed physics back to the commitment, so the theorem's "uniqueness" at these joints is uniqueness *relative to the observed inputs*, not derivation of them. This does not collapse the residue into tautology — the representation-theoretic consequences remain genuine content — but it locates the boundary precisely: the framework forces the consequences of the cubic lattice, and selects the cubic lattice from the world. The full three-way classification (forced / empirically selected / definitionally stipulated) is given in [SM §8.3, *Derivation status: the three tiers*]; the honest one-line statement of the framework's reach is that it is a maximally tight compression of observed physics onto the cubic-lattice commitment, not a derivation of physics from the bare fact of observation.

The structural assumptions A1–A6 exclude several broad classes of theory from the outset:

- Intrinsically continuum theories (violate A1)
- Intrinsically stochastic theories (violate A2)
- Theories with unbounded coupling degree (violate A3)
- Theories violating center independence, linearity, or background independence (violate A4–A6)

The framework's claim is that within the stated structural class the **local reconstructed residue** is unique up to gauge. The Bell-inclusive substratum is not proved unique. A separate argument would be required to rule out alternatives outside this class. The framework's defense of A1–A6 (in [SM §2] and elsewhere) is that they are natural given the empirical inputs E1–E7, but this defense is itself an argument, not a theorem.

Remark (Hypothesis dependencies). Each structural assumption can in principle be weakened or replaced:

- A1 (finiteness) is supported by E3; alternative would be an effective-finiteness argument with continuum UV completion, which the framework does not develop.
- A2 (determinism) is the key ontological commitment; stochastic alternatives would produce different reconstructions.
- A3 (bounded coupling) is conventional for physical systems; long-range couplings would give different dimensional structure.
- A4 (center independence) rules out preferred-frame theories.
- A5 (linearity) is the strongest restriction; it is equivalent to amplitude-scale gauge invariance (linearity-equivalence lemma, [SM §4.1]), so nonlinear wave equations on finite lattices are the separate class one obtains exactly when that gauge principle is dropped.
- A6 (background independence), as covariance under local internal-index transformations with the link coupling transformed alongside, is the standard gauge-theoretic requirement; the stronger fixed-background invariance, with the coupling held fixed, is a separate condition and is not what A6 asserts.

Weakening any A_i does not merely enlarge the equivalence class; it potentially produces *different* reconstructions that may not agree with observation. The framework's claim is that E1–E7 + A1–A6 is a consistent set of inputs producing our observed physics; alternative input sets are not investigated.

Remark (Continuum extension). The lattice-level predictions are the framework's primary claims — the lattice is the fundamental description, not an approximation to a continuum theory. The structural results (gauge group, representations, generation count) are algebraic/topological properties of the lattice theory and are exact. Quantitative observables (scattering amplitudes, mass ratios) are compared to experiment through continuum perturbation theory; the lattice-continuum discrepancy at any experimentally accessible energy E is suppressed by $(E\epsilon/\hbar c)^2 = 4\sigma(E/M_{\mathrm{Pl}})^2$, which at $\sigma = 1$ is $\sim 10^{-32}$ at LHC energies. The rigorous proof that lattice Yang-Mills defines a continuum limit with a mass gap (a Clay Millennium Prize problem) is an open problem in mathematics, but it is not required by the OI framework: the lattice theory is complete, and its predictions are lattice-exact.

The epistemic structure is as follows. A reasonable objection is that the framework treats the lattice as fundamental (so the mass-gap problem can be set aside) while also using continuum perturbation theory to extract the numbers that match data (which would make the continuum limit load-bearing). The position is consistent. The framework's claim is that the lattice theory is the complete physical content and its predictions are exact *at the lattice level*. The comparison to experiment is then a comparison between two objects: the lattice theory's prediction extracted via continuum PT (a controlled approximation with error bounded by $(E\epsilon/\hbar c)^2 = 4\sigma(E/M_{\mathrm{Pl}})^2 \sim 10^{-32}$ at $\sigma = 1$), and the continuum-QFT fit to the experimental data (the standard way the data is reported). The empirical agreement is therefore agreement between the lattice

theory's continuum approximation and the continuum-QFT data fit, with the approximation error in the first object ~ 30 orders of magnitude below the experimental precision of the second, so the lattice-vs-continuum gap is not the source of any observed agreement or disagreement at accessible energies. The mass-gap problem is not load-bearing for this: it concerns whether a continuum limit *exists as a rigorous mathematical object*, whereas the framework requires only that continuum PT is an accurate *calculational approximation* to the lattice theory's predictions at low energy — the standard status of lattice-to-continuum perturbative matching, which does not require the existence of the rigorous limit. The lattice is fundamental, the continuum is a calculational tool with a quantified error, and the experimental comparison is between the tool's output and continuum-reported data.

Remark (Finite observation sets vs. idealized empirical inputs). The empirical inputs E1–E7 are stated as structural facts about observed physics — the full apparatus of QM, Bell violations attaining Tsirelson's bound, the holographic entropy bound, exact rotational invariance, propagating gravitational waves, stable matter on all observed scales, and spatial-curvature concordance. Operationally, no finite observation set fully establishes any of these; finite measurements support them only in the sense of being consistent with the data so far. The reconstruction theorem holds for the empirical inputs *as stated* — the idealized infinite-data regime — and proves uniqueness of the local lattice/gauge residue under that regime; Bell-inclusive uniqueness remains open. Reconstruction from finite observation sets is a related but distinct question: with finite data, multiple equivalence classes may all be consistent, with the data discriminating among them only weakly. The framework's operational answer to the finite-data question is the cumulative weight of the quantitative predictions of [SM §7] — twenty-two classified retrodictions of gauge couplings, CKM/PMNS angles, mass ratios, and the Koide relation (two parameter-free structural and five conditional structural, the remainder layered, chained or fitted, the gauge couplings among the fitted), the measured comparisons each at $\lesssim 1\sigma$ — which collectively rule out reconstructions that would have produced different values. Theorem 23 establishes idealized local-sector uniqueness; the [SM §7] prediction set supplies the operational evidence for that local structural residue that a skeptical reader can check against data.

The reconstruction establishes a bidirectional correspondence for the **local reconstructed residue**:

$$\text{Observed local lattice/gauge data (E1–E7) + Structural assumptions (A1–A6)} \quad \longleftrightarrow \quad [(S,\varphi)]_{\mathrm{loc}}/\mathcal{G}_{\mathrm{sub}}.$$

The local lattice/gauge structure and the corresponding observed inputs determine each other up to the stated gauge equivalence. Bell-inclusive completion data sit outside this arrow: their existence is conditional on H-Bell and their uniqueness is open. The distinction between "mathematics describes reality"

and "mathematics is reality" has no empirical content for any embedded physical observer — it is itself gauge in the framework's specific physics-internal sense, provably undecidable by any measurement (cf. Theorem 24). The framework does not claim the question is absolutely meaningless; whether it has substantive content in mathematical foundations or other modes of inquiry not bound by embedded physical observation is a separate matter the framework does not address. This reframes Wigner's puzzle: the "unreasonable effectiveness" of mathematics is a theorem modulo the structural assumptions, not a mystery — the reconstruction theorem supplies, modulo gauge, the specific mathematical structure that physics is observationally equivalent to.

Remark (ER=EPR at substratum and emergent levels). The Maldacena-Susskind ER=EPR conjecture [3] proposes that any two entangled systems are connected by an Einstein-Rosen bridge, with the strong form asserting that entanglement *is* a non-traversable wormhole — the same ontological structure under two descriptions. The framework here engages this conjecture at two distinct levels.

At the **emergent level**, the conjecture's *weak* form is reproduced: any pair of entangled subsystems in the emergent QM description shares boundary modes via the partition trace-out, and the holographic dictionary maps shared boundary modes to a connecting geometric structure in the emergent gravitational description. This recovers the ER=EPR weak form — entangled systems are connected by a wormhole-like geometric object in the emergent theory — as a structural consequence of the framework's holographic architecture rather than as an independent postulate.

At the **substratum level**, the conjecture's *strong* form (entanglement is identical to wormhole geometry, not merely correlated with it) is predicted *false* at the representative level. Two distinct partition representatives can produce identical emergent EPR correlations while having different substratum representative structures — there is no faithful bijection from EPR pairs to substratum geometric features. The strong identity holds at most up to gauge equivalence on the substratum side. Whether a gauge-orbit reformulation of strong ER=EPR — "EPR-equivalence-classes correspond to ER-equivalence-classes" — is true in the framework remains open; the test would be whether the substratum's gauge group $\mathcal{G}_{\text{sub}}$ acts transitively on the set of substratum representatives that produce a given EPR correlation pattern.

This produces a sharp empirical signature: the framework's operational predictions agree with weak ER=EPR (which would also be predicted by AdS/CFT, by quantum error correction in holography, and by other emergent-geometry approaches) but disagree with strong ER=EPR at the representative level. Distinguishing these experimentally would require probing substratum-scale features, which is currently beyond reach. The disagreement is therefore a structural prediction rather than a testable one at present, but it concretely positions the framework relative to the broader ER=EPR program.

4. The Substratum Gauge Group

The equivalence relation $\sim$ in the reconstruction theorem has a precise structure. Define: $(S, \varphi) \sim (S', \varphi')$ if the two systems produce identical emergent physics — the same transition probabilities $T_{ij}(t)$, the same emergent Hamiltonian (up to D-gauge), and the same $\hbar$ — for all partitions of the same structural class. The set of transformations mapping (S, φ) to an equivalent (S', φ') is a group — the *substratum gauge group* $\mathcal{G}_{\text{sub}}$.

Theorem 24 (Generators of the substratum gauge group). *$\mathcal{G}_{sub}$ contains the independent families of transformations (i), (iii) and (iv) below, and, under the alphabet-size hypothesis of [SM, §2.7], the family (ii):*

(i) State relabeling. For any bijection $\pi : S \to S$, the conjugate $(S, \pi \circ \varphi \circ \pi^{-1}) \sim (S, \varphi)$. The transition probabilities depend on the coupling structure of φ, not on which labels are attached to which states. This is an $|S|!$-element subgroup — vastly larger than any gauge group in the Standard Model. *(Scope.)* The preserved observables here are the coupling-graph invariants of G_φ; a site-wise *nonlinear* relabelling of the alphabet preserves G_φ but alters the field-amplitude dispersion $\omega(k)$ of [SM §4.1], so it preserves observables at the graph level but not the emergent Hamiltonian. The relabellings preserving $\omega(k)$ as well are exactly the additive-automorphism (affine) ones — the amplitude-scale subgroup (Remark below).

(ii) Alphabet change (under the alphabet-size hypothesis). Replacing the local state space $\mathbb{Z}/q\mathbb{Z}$ with $\mathbb{Z}/q'\mathbb{Z}$ for any $q' \geq 2$, while preserving the coupling graph and dynamics class, leaves all observables unchanged if the alphabet-size hypothesis of [SM, §2.7] holds; [SM, §2.7] lists the results proved for every $q \geq 2$ and those that carry q. This family is parametrized by all integers $q \geq 2$.

(iii) Deep-sector enlargement. Adjoining to $\mathcal{C}_D$ (the deep hidden sector beyond the boundary layer) additional degrees of freedom that no dynamics couples to $\mathcal{C}_V \times \mathcal{C}_B$ — so that the bijection on the enlarged space is $\varphi \times \varphi_{\text{new}}$ for a bijection φ_{new} of the adjoined degrees of freedom — leaves every $T_{ij}(t)$ unchanged at every t, and with it the emergent description; removing such degrees of freedom is the inverse. The deep sector may be finite of any size, or infinite. *(Scope.)* An enlargement whose dynamics couples to the boundary layer, even slowly ($\tau_B^D \gg \tau_S$), is not in this family: the boundary-only dependence lemma [GR, §3.2] gives $T_{ij}(t) = T_{ij}^{(B)}(t) + \mathcal{O}(t/\tau_B)$, so such an extension preserves the leading-order transition law $T^{(B)}$ on accessible timescales $t \ll \tau_B$, not the transition probabilities themselves.

(iv) Graph isomorphism (up to statistical isotropy), with a qualification on the gauge sector. Two coupling graphs G_φ and $G_{\varphi'}$ that are quasi-isometric with the same polynomial growth exponent d, the same spectral mode count, and the same statistical isotropy at large scales produce the same emergent *quantum-*

mechanical, dimensional, and gravitational content (Theorems 1–6); for this content the regular cubic lattice $\mathbb{Z}^3$ and any bounded-degree random graph with $d = 3$ polynomial growth and statistical isotropy are gauge-equivalent. The *gauge-group* content of Stage 2(d) is where this crystallography becomes physical: the multiplicities $(3, 2, 1)$ are fixed by the octahedral point group acting on the six link directions (Theorem 7; [SM §4.5–4.6]), and the local point group is *not* a quasi-isometry invariant — only the spectral mode count $2d_s = 6$ transfers to a generic isotropic graph, not the cubic decomposition. For the gauge-group content, therefore, the gauge freedom in (iv) is the narrower one of rigid orientation together with isomorphisms preserving the cubic (octahedral) local structure; the simple-cubic Bravais type itself is an empirically-selected input (§3, Refinement below; [SM §8.3]), not a gauge choice. A generic isotropic random graph reproduces the emergent-QM content but not the Standard Model gauge group. The gauge group itself is a universality-class invariant, shared with any Standard-Model-reproducing class through different machinery ([Structure §2.3, §14.4]); within OI's class it is realized through the cubic-group commutant (§4.6).

Proof. Each generator preserves all inputs to the derivation chain. (i): conjugation preserves the coupling graph G_φ up to relabeling, hence all graph-dependent quantities (area law, dispersion, dimension, eigenvalue multiplicities). (ii): this is the alphabet-size hypothesis of [SM, §2.7], under which the family is stated; the results [SM, §2.7] proves for every $q \geq 2$ are preserved without it. (iii): the visible marginal of $\varphi \times \varphi_{\text{new}}$ averages over the adjoined coordinates a quantity that does not depend on them, so every $T_{ij}(t)$ is unchanged exactly, whatever the size of the adjoined sector, and the emergent Hamiltonian and $\hbar$, built from the same visible data and the same $\mathcal{C}_V \times \mathcal{C}_B$ dynamics, are unchanged with them. (iv): the emergent-QM/dimension/gravity derivation chain (Theorems 1–6) uses only statistical properties of G_φ (dimension via Myrheim-Meyer, isotropy, bounded degree), not the specific graph; the gauge-group derivation of Stage 2(d) additionally uses the octahedral point-group action on the six link directions, which the simple-cubic representative supplies and a generic isotropic graph does not, so for that content (iv) is restricted to the orientation- and octahedral-structure-preserving isomorphisms (see the qualification in the statement). □

Corollary (effective finiteness and gauge-class transfer). Three statements make A1's two-part status precise. *(i) The bound.* E3, read as a Hilbert-space dimension cutoff, bounds the observationally relevant space: $|\mathcal{C}_V \times \mathcal{C}_B| \leq e^{S_\partial}$ with $S_\partial = A/4$ in Planck units — for the de Sitter horizon, $e^{10^{122}}$. This dimension-cutoff reading is the interpretive premise of A1, and this is the only place it enters. *(ii) Boundary-only dependence.* For accessible times $t \ll \tau_B$, $T_{ij}(t) = T_{ij}^{(B)}(t) + \mathcal{O}(t/\tau_B)$ ([GR §3.2]), so at leading order every emergent observable on accessible timescales is a functional of $\mathcal{C}_V \times \mathcal{C}_B$ data alone. *(iii) Transfer.* Along the orbit of (S, φ) under generator (iii), every $T_{ij}(t)$ is unchanged at every t, so every statement expressible through $\{T_{ij}(t)\}$, recurrence-scale statements

included, holds on one member if and only if it holds on every member, finite or infinite. An extension coupled to the boundary layer with $\tau_B^D \gg \tau_S$ alters $\mathcal{C}_D$ and its dynamics while leaving $T^{(B)}$ unchanged, so the two systems share only the leading-order law: a statement expressible through $\{T_{ij}^{(B)}(t) : t \ll \tau_B\}$ transfers by (ii), while a statement about the transition probabilities themselves transfers only if it is insensitive to $\mathcal{O}(t/\tau_B)$ corrections. A transition probability equal to zero on one system and to $\varepsilon = \mathcal{O}(t/\tau_B) > 0$ on the other is the elementary statement that does not transfer. P-indivisibility at the recurrence scale — established on finite representatives by recurrence ($\varphi^N = \mathrm{id}$, [Main §2.3]) — is not an accessible-timescale statement: across coupled extensions it holds on the finite members, and on infinite-deep-sector members, where recurrence is unavailable, the accessible-timescale backflow lemma ([Main §2.3]) gives the accessible-window property it states, a different and weaker conclusion. A1's choice of the finite representative is therefore a convenience of proof for content carried by the leading-order accessible-timescale law; recurrence-scale content is stated for finite representatives. □

Remark (amplitude-scale gauge — the unobservability of absolute field scale). One gauge freedom invoked elsewhere is named here as its single authoritative home, and is best stated *operationally*: an embedded observer has no access to the absolute scale of the substratum field value — only to the emergent transition probabilities and coupling structure (§2.7). This is the discrete analogue of the field-normalization redundancy of ordinary field theory, where rescaling a field changes nothing physical because only correlators, not absolute amplitudes, are observable. It is realized as *amplitude-scale invariance*: the rescalings $\phi \mapsto \lambda\phi$ (λ a unit) and shifts $\phi \mapsto \phi + c$ — the additive-automorphism (affine) relabellings of the alphabet — leave every observable unchanged. These affine relabellings are exactly the site-wise relabellings that preserve not only the coupling graph but the field-amplitude dispersion $\omega(k)$ (the emergent Hamiltonian), so amplitude-scale gauge is one instance of the framework's master principle that *observationally indistinguishable transformations are gauge* — the same principle behind generators (i)–(iv) and the q-size freedom of §2.7, not an ad hoc addition. Two derivations invoke it: the linearity equivalence of [SM §4.1] (amplitude-independent dispersion $\iff$ affine dynamics) and the $SU(N)$ reduction of Stage 2(d) above; its status as an operational input rather than a theorem — and the sense in which those two results are conditional on it — is set out in the *Status* note immediately below.

Status (open, referee-grade). Amplitude-scale gauge is adjoined as an explicit operational principle: it is not one of (i)–(iv) — generator (i) over-counts (its full $|S|$! includes dispersion-altering nonlinear relabellings, which would vacate rather than establish linearity) and (ii) is alphabet-*size* only — so the results invoking it are conditional on it. Stated operationally it is *not* viciously circular: the input (absolute amplitude is unobservable) is strictly weaker than the output (additive dynamics), and a nonlinear F on the same alphabet is a coherent object the principle *excludes* rather than presupposes. The one route that would

make linearity *unconditional* — deriving the unobservability of absolute amplitude from the observer architecture itself — is by contrast circular: what the embedded observer can resolve through the trace-out is dynamics-dependent (a nonlinear F induces amplitude-dependent emergent dispersion, which *is* observable), so "absolute amplitude is unobservable" already presupposes linearity. Amplitude-scale gauge therefore has the status of an operational input on a par with center independence and isotropy, not a theorem; whether it is independently warranted by the observer architecture is the question flagged for external review.

Formalization status of the linearity step (resolved as a sharpened axiom, not a closure). A dedicated audit of this step settles its logical status, and the outcome is worth stating precisely because it is the prototype for how the framework's "open" foundational steps resolve. The linearity-equivalence lemma — *all first finite-differences of the update rule over $\mathbb{Z}/q\mathbb{Z}$ are base-point-independent iff the rule is affine as a function* — has been verified exhaustively for $q \leq 6$ in one neighbour-coordinate and $q \leq 3$ in two, and on 2.5×10^5 random rules at larger (q, m), with zero counterexamples; together with the telescoping proof (constant single-coordinate differences force $F(x) = F(0) + \sum_i c_i x_i$, well-defined on the torus since $qc_i \equiv 0$) this is a complete elementary result, not a genericity heuristic. The question of whether the *q-size* gauge freedom of [SM §2.7] *entails* amplitude-scale gauge resolves in the negative, and sharply: the weak reading of §2.7 (size-independence of predictions) does not reach rescaling at fixed q; the strong reading (arbitrary relabellings of $\mathbb{Z}/q\mathbb{Z}$ are gauge) *overshoots* — a nonlinear permutation conjugating a linear rule preserves cycle structure (hence emergent transition probabilities up to relabelling) while being nonlinear, so it would render a nonlinear rule gauge-equivalent to a linear one and thereby *destroy* linearity-necessity rather than establish it (verified explicitly on $\mathbb{Z}/5\mathbb{Z}$). The principled object that delivers exactly what is needed without overshooting is the **additive-automorphism** group $x \mapsto \lambda x + c$ (λ a unit) — a proper subgroup of all permutations that preserves the alphabet's "+", hence the dispersion. The upshot, in the framework's three-tier classification ([SM §8.3]): linearity is **not** promoted to a forced (Tier-1) consequence; it is a Tier-3 *definitional stipulation*, but a sharpened one — the bare assumption "the dynamics is linear" is replaced by the more primitive and better-motivated assumption "the alphabet's additive automorphisms are gauge," from which linearity follows as a certified theorem. This is the characteristic resolution shape for the framework's foundational gaps: pressing for a proof does not close the gap to a derivation but relocates the irreducible assumption to a more primitive and more defensible place, and names it exactly.

Completeness of the generators (conditional). The four families exhaust $\mathcal{G}_{\text{sub}}$ conditional on Lemma 24.1's semigroup-transfer step (below), family (ii) entering under the alphabet-size hypothesis. The argument proceeds by showing that any observables-preserving transformation decomposes into (i)–(iv). The argument has two logical parts that are worth separating, because the completeness (as opposed to soundness) direction is the substantive one: soundness —

that each of (i)–(iv) preserves observables — is established above; completeness — that *every* observables-preserving transformation lies in the group generated by (i)–(iv) — is what follows.

Lemma 24.1 (Uniqueness of the bijection-generated dilation). *Let Φ_t ($t \in \mathbb{Z}_{\geq 0}$) be the visible-sector channel family induced by a substratum bijection φ on $S = \mathcal{C}_V \times \mathcal{C}_H$ with uniform prior on $\mathcal{C}_H$, via $\Phi_t(\rho_V) = \mathrm{Tr}_H[\hat{\varphi}^t\,(\rho_V \otimes \omega_H)\,\hat{\varphi}^{-t}]$, where $\hat{\varphi}$ is the permutation unitary of φ on $\mathbb{C}^S$ and $\omega_H = \mathbb{1}/|\mathcal{C}_H|$. Suppose a second substratum φ' on $S' = \mathcal{C}_V \times \mathcal{C}'_H$ induces the identical family $\Phi'_t = \Phi_t$ for all t. Then there is a partial isometry $W : \mathbb{C}^{\mathcal{C}_H} \to \mathbb{C}^{\mathcal{C}'_H}$ such that $\hat{\varphi}'$ and $\hat{\varphi}$ agree on the boundary sector through conjugation by $\mathbb{1}_V \otimes W$, up to enlargement/reduction of the deep sector $\mathcal{C}_D \subset \mathcal{C}_H$ that couples to $\mathcal{C}_V$ at no order in t.*

Proof structure. The family $\{\Phi_t\}$ is not an arbitrary collection of channels: it is the orbit of a single unitary $\hat{\varphi}$, i.e. $\Phi_t = \mathcal{E} \circ \mathrm{Ad}_{\hat{\varphi}}^t$ restricted to the visible factor, where $\mathrm{Ad}_{\hat{\varphi}}(X) = \hat{\varphi} X \hat{\varphi}^{-1}$. Three steps.

(1) *The dilation is a single unitary, not a per-time family.* Because every Φ_t is generated by the same $\hat{\varphi}$, specifying the family is equivalent to specifying the pair $(\hat{\varphi}, \omega_H)$ up to the freedom that leaves the visible marginal invariant at every t. This reduces the completeness question for the family to a uniqueness question for one unitary dilation of the discrete one-parameter semigroup $t \mapsto \Phi_t$, eliminating the "independent isometry per time" loophole that a channel-by-channel argument would leave open.

(2) *Minimal dilation is unique up to environment unitary (GNS/Stinespring).* Restrict to the cyclic subspace $\mathcal{C}_B = \overline{\mathrm{span}}\{\hat{\varphi}^t(\mathcal{C}_V \otimes \mathrm{supp}\,\omega_H)\}$ — the boundary sector reachable from the visible factor under the dynamics. On $\mathcal{C}_V \otimes \mathcal{C}_B$ the dilation is minimal by construction. The Stinespring/GNS uniqueness theorem for a unitary-generated dilation states that two minimal dilations of the same channel semigroup, on the same visible space, are related by a unitary isomorphism of the environment that intertwines the dilating unitaries. Applying it to $\hat{\varphi}$ and $\hat{\varphi}'$ on their respective boundary sectors yields a unitary $W_B : \mathcal{C}_B \to \mathcal{C}'_B$ with $\hat{\varphi}' = (\mathbb{1}_V \otimes W_B)\,\hat{\varphi}\,(\mathbb{1}_V \otimes W_B)^{-1}$ on $\mathcal{C}_V \otimes \mathcal{C}_B$.

(3) *The deep sector is the non-minimal remainder.* Any $\mathcal{C}_H$ larger than $\mathcal{C}_B$ decomposes as $\mathcal{C}_B \oplus \mathcal{C}_D$ with $\mathcal{C}_D$ decoupled from the visible factor at every order in t. This exact decoupling is the deep sector of the lemma's statement and of generator (iii); the boundary-only dependence lemma [GR §3.2] does not supply it, since that lemma bounds the effect of a slowly coupled deep sector at $\mathcal{O}(t/\tau_B)$. The non-minimal dilation freedom of Stinespring is exactly this decoupled remainder, which is generator (iii). Composing, $W = W_B$ extended by any partial isometry on $\mathcal{C}_D$ gives the claimed relation. □

What a specialist must verify in Lemma 24.1. The load-bearing step is (2): that the standard Stinespring/GNS uniqueness-up-to-environment-unitary, nor-

mally stated for a single CP map, transfers to the discrete unitary-generated semigroup $\{\Phi_t\}$ on the cyclic boundary sector. This is expected to hold because a single generating unitary makes the dilation a genuine GNS object (a cyclic representation of the $*$-algebra generated by $\hat{\varphi}$), for which the uniqueness theorem is standard; but the precise statement — that minimality on the cyclic subspace plus equality of the full family forces the intertwining unitary — is the point at which an operator-algebra specialist should confirm the argument. The framework asserts (2) as following from the cyclic-representation structure; it does not here reproduce the GNS uniqueness proof, and (2) is the step to check. Steps (1) and (3) are bookkeeping given (2). For the specialist, the relevant precedent is that the single-map uniqueness (minimal Stinespring dilations are unique up to a unitary on the environment — the standard uniqueness clause of Stinespring's dilation theorem) is known to transfer to *iterated* and *discrete-semigroup* dilations under a minimality condition: the iterated-channel construction on a cyclic subspace uses exactly "the Stinespring dilation of the repeated map, with uniqueness up to unitary equivalence following from uniqueness of the Stinespring representation and cyclicity following from minimality" (as in the quantum-spin-chain finitely-correlated-state literature, e.g. Bratteli–Jorgensen–Kishimoto–Werner-type arguments; arXiv:math-ph/0505035), and minimal dilations of discrete-semigroup actions are canonically unique under minimality (Laca, *From endomorphisms to automorphisms and back*, arXiv:math/9911135). The step (2) claimed here is an instance of this pattern specialised to the semigroup $\{\Phi_t\}$ generated by the single unitary $\hat{\varphi}$; the specialist's task is to confirm the specialisation, not to establish an unprecedented result.

Step 1: the visible channel. By assumption g preserves the full time-resolved family of visible-sector CPTP channels $\{\Phi_t\}_{t\geq 0}$, not merely a single channel; this preserves $T_{ij}(t)$ for all t. Reconstruction from transition probabilities is conditional on H-BG and the dimension, gap and overlap hypotheses of [GR §3.3], and its alternatives include an energy shift and the antiunitary reflection as well as diagonal rephasing. If the common channel family is exactly unitary, equality of the conjugation actions gives a stronger conclusion: the generators differ only by an energy shift ([GR] Corollary A.3). For general CPTP dynamics a unique Hamiltonian part is not supplied by this argument. The observable tuple defining $\mathcal{G}_{\mathrm{sub}}$ separately requires preservation of H up to the specified gauge and of $\hbar$.

Step 2: the dilation freedom. Fixing $\{\Phi_t\}$ does not fix the substratum: the substratum is a dilation of the channel family. Here the quantifier direction must be handled with care. Stinespring's theorem provides, for a single channel, that any two minimal dilations on a common output space differ by a unitary on the dilating (hidden) space, and any two dilations differ by a partial isometry once non-minimality is allowed. The completeness argument requires the converse-facing statement: that the *only* substratum-level freedom consistent with fixed $\{\Phi_t\}$ is dilation freedom of this Stinespring type plus the structural freedoms below. This holds because the substratum is, by construction (§2, [Main §2]), nothing more than a dilation datum: a finite set, a bijection, and a partition into

visible and hidden sectors. Any g preserving $\{\Phi_t\}$ acts on this datum and must therefore (a) preserve the visible factor up to the relabeling that does not affect Φ_t — generator (i) restricted to $\mathcal{C}_V$ followed by the D-gauge already quotiented in Step 1; (b) act on the hidden factor by a transformation that leaves every Φ_t invariant, which by Lemma 24.1 is a hidden-sector partial isometry — generator (i) restricted to $\mathcal{C}_H$ when cardinalities match, or generators (i)+(iii) when they differ; and (c) possibly change a deep sector that no dynamics couples to the boundary layer, which cannot affect any Φ_t at all — generator (iii). A change to deep-sector dynamics coupled to the boundary layer, however slow, alters Φ_t at $\mathcal{O}(t/\tau_B)$ ([GR §3.2]); it preserves the leading-order law $T^{(B)}$ but not the family $\{\Phi_t\}$, and so lies outside the transformations this argument classifies. The use of the full family $\{\Phi_t\}_{t\geq 0}$ rather than a single channel is what closes the gap a single-channel Stinespring argument would leave: a transformation preserving Φ_{t_0} at one time but altering the generator would fail to preserve Φ_t at other times, and is therefore excluded by the hypothesis that *all* $T_{ij}(t)$ are preserved.

Step 3: the structural freedoms. Two substrata may induce the same $\{\Phi_t\}$ while differing in their local state-space size (alphabet) or in the specific coupling graph realizing the dynamics. Under the alphabet-size hypothesis the alphabet may change with no effect on any observable (generator (ii), [SM §2.7]); an alphabet change that alters some observable is not in $\mathcal{G}_{\mathrm{sub}}$. For the emergent-QM/dimension/gravity content the coupling graph may be replaced by any graph in the same statistical class — same growth dimension, spectral dimension, and large-scale isotropy — since that part of the derivation chain reads only these statistical invariants of G_φ (dimension via Myrheim–Meyer, isotropy, bounded degree) and never the specific graph (generator (iv)). The gauge-group content of Stage 2(d) reads out the octahedral structure, as noted in Theorem 24(iv): it uses the octahedral point group on the six link directions, so there the admissible replacements are restricted to graphs preserving that local structure; the simple-cubic type is an empirically-selected input.

Conclusion. Steps 1–3 enumerate the complete set of substratum-level data on which $\{\Phi_t\}$, H, and $\hbar$ depend: the visible-hidden dilation structure (governed by (i), (iii)), the deep sector (governed by (iii)), the alphabet (governed by (ii) under the alphabet-size hypothesis), and the coupling-graph statistical class (governed by (iv)). No further substratum freedom exists, because the substratum *is* the dilation datum and these are its parts. Hence any observables-preserving g lies in $\langle$(i),(ii),(iii),(iv)$\rangle$, and the four families generate $\mathcal{G}_{\mathrm{sub}}$. □

Scope of the completeness claim. The argument establishes completeness relative to the observable set $\{T_{ij}(t)$, H up to D-gauge, $\hbar\}$ and the substratum ontology of §2 (finite set, bijection, visible/hidden partition). It does not claim completeness against a richer substratum ontology carrying structure beyond the dilation datum — e.g., a substratum equipped with additional intrinsic fields not reflected in any Φ_t. Within the framework such additional structure is by definition unobservable and hence already gauge; a referee adopting a different substratum ontology would need to re-examine Step 2 accordingly. The

completeness claim is therefore precise rather than absolute — and conditional: subject to Lemma 24.1, the four families exhaust the observables-preserving transformations of the dilation-datum substratum, which is the object the reconstruction theorem reconstructs.

Three candidate fifth families are explicitly subsumed:

- *Time reversal* ($\varphi \to \varphi^{-1}$): the wave equation's T-invariance gives $\varphi^{-1} = T \circ \varphi \circ T^{-1}$ where T is the phase-space layer swap $(x(t), x(t+1)) \mapsto (x(t+1), x(t))$. This is generator (i) with $\pi = T$.
- *Hidden-sector dynamical reparametrization* (beyond enlargement): by Stinespring uniqueness, any two same-channel dilations of equal hidden-sector dimension differ by a hidden-sector unitary — generator (i) restricted to $\mathcal{C}_H$.
- *Visible-sector emergent global phase* ($U \to e^{i\theta}U$): at the substratum level S is a finite set with no complex structure; the emergent phase is trivially the identity on (S, φ).

The gauge hierarchy. Three levels of gauge symmetry appear in the framework, each projecting onto the next through the trace-out:

Level 3 (substratum): $\mathcal{G}_{\text{sub}}$ acts on (S, φ) before the trace-out. It is the largest gauge group and includes transformations with no analog in the emergent description (deep-sector enlargement, and alphabet change under the alphabet-size hypothesis).

Level 2 (emergent QFT): $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ is the commutant of the coupling matrix M, acting on the emergent fields. It is the image of $\mathcal{G}_{\text{sub}}$ restricted to transformations that permute internal components within the eigenspaces of M.

Level 1 (emergent Hamiltonian): The D-gauge $H \to DHD^\dagger$ with D a diagonal unitary, acting on the emergent Hamiltonian within the emergent QM. It preserves every transition probability exactly. Energy shifts and the antiunitary reflection are also invisible; completeness of these alternatives under the generic gap and overlap hypotheses is conditional on H-BG ([GR §3.3]).

Each level is contained in the one above: Level 1 $\subset$ Level 2 $\subset$ Level 3. The trace-out projects Level 3 onto Level 2 (the SM gauge group is the shadow of $\mathcal{G}_{\text{sub}}$ visible to the emergent QFT), and restricting to the Hamiltonian projects Level 2 onto Level 1.

Remark. The substratum gauge group is not a symmetry of a Lagrangian or an action — no Lagrangian exists at the substratum level. It is a symmetry of the *equivalence class of substrata*, defined by the condition that all observables are preserved. The emergent gauge symmetries (Levels 1 and 2) are Lagrangian symmetries in the standard sense, derived from the substratum through the trace-out.

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5. Synthesis: Three Projections of (S, φ)

The reconstruction theorem (§3) establishes that the stated empirical and structural inputs, together with M1-T, C2-mixing and Lemma 24.1's completeness step, uniquely fix the **local propagating lattice/gauge residue** modulo the stated substratum-gauge freedoms. M1-B and H-Bell separately condition the existence of a measurement-independent Bell-violating completion; Bell-inclusive uniqueness is open. The substratum gauge group (§4) makes the equivalence relation precise and identifies the Standard Model gauge group as the visible-sector shadow of $\mathcal{G}_{\text{sub}}$. This section makes the synthesis claim explicit: quantum mechanics, general relativity, and the arrow of time are not three independent theories but three projections of the same finite deterministic construction.

5.1 The three projections

The framework's derivations apply the same trace-out machinery to the same (S, φ) along three projections: a visible-sector projection that produces the reduced description admitting the quantum representation ([Main]) and the Standard Model ([SM]); a boundary-thermodynamic projection that produces general relativity ([GR]); and an arrow-of-time projection (§5.4) that produces the emergent cosmological, thermodynamic, memory, and measurement arrows from a single structural clock and the C1–C4 observer-selection theorem of [Main, §4.6].

Visible-sector projection. At the level of the embedded observer's epistemic access, the trace-out over the hidden sector produces the exact finite observable-law correspondence established in [Main]: $S \iff D \iff Q_{\text{fb}}$. The fixed-basis unitary/Born representation is therefore exact and bidirectional but universal. The nontrivial OI content is history readback: C4 forces conditional memory and hidden predictive structure in every faithful completion; C1 and C3 diagnose the mediation and its capacity, while C2-structural supplies persistence to readback. In the spatial theory the dependency graph additionally supplies exact causal cones for finite-range couplings. Full equality with the standard coherent instrument/composite theory remains the narrower operational-lifting statement of [Main §3.4]. Bell is separate: the measurement-independent deterministic completion reaches quantum Bell violation only by ontic parameter dependence, with operational no-signaling [Main §3.3]. Applied to a cubic lattice with the wave equation as substratum dynamics — itself uniquely selected among second-order reversible nearest-neighbor dynamics by center independence, isotropy, and linearity ([SM, §4.1]) — this projection produces the Standard Model gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ (as the commutant of the coupling matrix, [SM, §§4.4–4.6]), three generations (physical reading conditional on H-spin', [SM §4.7]; chirality on H-χ', [SM §4.8]) and the Higgs as a composite scalar in the singlet staggered taste ([SM, §4.7], Theorems 8–11), and anomaly-free hypercharges ([SM, §4.9], Theorems 14–15). Twenty-two quantitative predictions follow ([SM, §7]), including the Cabibbo angle from a single

Brillouin-zone distance, the Koide angle from a cubic-group quadratic Casimir, all three PMNS angles within 1.1σ (vs NuFIT 6.0), six fermion masses from one empirical input to better than 1%, and all three SM gauge couplings at M_Z ([SM, §6.3]).

Boundary-thermodynamic projection. At the level of the partition boundary, classical horizon thermodynamics — the temperature, entropy, and dynamical evolution of the causal horizon as a gravitating object — produces general relativity through Jacobson's thermodynamic argument, conditionally on the open covariant step ([GR, §8.7]): the Clausius relation $dE = T\,dS$ applied at local causal horizons yields Einstein's equations with the same structural inputs that give the visible-sector projection its form. Applied to the cosmological horizon as a causal partition ([GR, §2]), this projection determines the emergent action scale $\hbar = c^3\epsilon^2/(4G\sigma)$ from thermal self-consistency ([GR, §3]), fixes the discreteness scale $\epsilon = 2\sqrt{\sigma}\,l_p$, for an entropy σ per boundary mode ($\epsilon = 2l_p$ at one nat per mode), as the unique simultaneous solution ([GR, §4]), recovers the Bekenstein-Hawking entropy $S = A/(4l_p^2)$, the mode count reproducing it once the calibration fixes $\epsilon/\sqrt{\sigma} = 2l_p$ and the 1/4 being the ratio $2\pi/8\pi$ of the KMS and first-law normalizations ([GR, §5]) — the action scale, the discreteness scale and the coefficient conditional on H-slope with the horizon and frame conditions of [GR, §2, §8.5]; GW250114 confirms the classical area theorem, the coefficient itself untested directly — and addresses the cosmological constant problem by identifying the quantum vacuum energy and the effective Λ as properties of logically distinct levels of description rather than commensurable quantities whose 10^{122} ratio requires cancellation ([GR, §6]), the descriptive distinction established and the dissolution conditional on the open coupling step ([GR, §6.2]). The framework carries dark energy in running-vacuum form — the functional form the local expansion shared by every smooth model, the content the Type II channel with magnitude $|\nu| \sim 10^{-32}$ (not derived at theorem level) — consistent with the Bertini et al. (2025) direct DESI-DR2-era fit at 2σ (which is consistent with the ΛCDM limit); a ~95% dark sector corollary as a structural feature rather than a new particle species; and a MOND acceleration scale $a_0 = cH/6$ from entropy displacement at the boundary ([GR, §7]).

The three projections share the same source ((S, φ)), the same trace-out machinery (marginalization over the hidden sector under conditions C1–C4), and the same structural inputs (the partition geometry, the boundary entropy, the substratum gauge group). They differ only in which structure of the embedded observer's description they derive: the visible-sector projection (§5.1, §5.3) works on the bulk dynamics and produces the quantum-representable reduced description and the Standard Model; the boundary-thermodynamic projection (§5.1) works on the classical thermal data at the partition boundary and produces general relativity; the arrow-of-time projection (§5.4) works on the horizon-clock parameter along observer worldlines and produces the cascade of emergent arrows.

The projections and the Poincaré structure of the partition. The three

projections are organized by how the observer-worldline-centered partition treats the Poincaré group, which makes the boost/rotation result of [Main §3.5] one facet of a larger pattern. The partition selects a preferred event — a spatial origin and a rest frame, jointly the observer's worldline — so the compact rotations about that worldline preserve the visible/hidden factorization ([Main §3.1]) and descend to exact spatial isotropy at the visible-sector level, while the noncompact generators are broken. Boosts, tied to the selected rest frame, surface as the cosmic rest frame (operationally the CMB frame) and the boost-sector Lorentz residual ([SM §3.1]); spatial translations, which merely relabel the observer's origin, are broken only observer-relatively and are recovered as spatial homogeneity — the equivalence of all worldline-centered descriptions; and time translation is broken intrinsically by the growth of the horizon, which is the boundary-thermodynamic and arrow projections (§5.4, [GR]). The partition's symmetry-breaking signature — isotropy preserved, a preferred rest frame, spatial homogeneity, a distinguished cosmic time — is the symmetry structure of a Friedmann-Robertson-Walker cosmology. The framework does not derive that structure (spatial isotropy enters as an empirical input); it exhibits it as the necessary shadow of worldline-centered observation, with ordinary momentum conservation holding in the visible sector up to corrections of order (length / horizon radius).

5.2 The QM-GR incompatibility as a category error

The conventional formulation of the QM-GR unification problem treats quantum mechanics and general relativity as two competing theories in the same logical category — two attempts to describe the same regime — and asks how to merge them into a single mathematically consistent framework. Every program of unification proceeds from this assumption: quantum gravity programs treat the metric as a quantum field to be quantized; geometric programs treat the wave function as a structure on the spacetime manifold; emergent programs treat one of the two as derived from a deeper substrate that recovers the other. The persistent failure of all three approaches to produce a quantitatively confirmed unification suggests that the underlying assumption is wrong.

In the present framework, the assumption is wrong because quantum mechanics and general relativity occupy different positions in the trace-out hierarchy. They are not two theories of the same regime at all — they are two projections of the same construction, viewed at different levels. The visible-sector projection that produces the quantum-representable reduced description and the boundary-thermodynamic projection that produces GR are not in tension because they refer to *different objects in the construction*. The wave function is a property of the embedded observer's compressed description of the visible sector; the metric is a property of the classical dynamics of the partition boundary. These are not two descriptions of the same physical system at different levels of approximation. They are descriptions of two different substructures of the same total object (S, φ).

The analogy that captures the structure most clearly is the one between a thermodynamic and a statistical-mechanical description of a gas. The thermodynamic description deals with pressure, temperature, and entropy; the statistical-mechanical description deals with particle positions, momenta, and microstate counting. These are not two competing theories of the gas, and the question "how do we unify thermodynamics and statistical mechanics?" is not a coherent question — they describe different things about the same system, and the relationship between them is one of projection (statistical mechanics produces thermodynamic quantities by averaging) rather than unification. The QM-GR relationship in the present framework is structurally similar: GR describes the thermodynamic (boundary, classical, deterministic) projection of (S, φ), and QM describes the statistical (visible-sector, compressed, stochastic-emergent) projection. The two are related by a definite procedure (the trace-out under conditions C1–C4), and the question "how do we unify them?" is dissolved rather than answered.

This is not the same as saying that quantum mechanics is "more fundamental" than general relativity, or vice versa. Both are emergent. Both depend on the trace-out and the partition. Both are derived rather than fundamental. The fundamental object is (S, φ) — a finite set with a deterministic bijection — and neither QM nor GR is a feature of (S, φ) itself. They are features of how an embedded observer or a horizon-bounded thermodynamic regime sees (S, φ) from inside. (Nor is the ranking question recoverable from inside: by the factoring argument ([GR §6.3]), no criterion grounded in the accessible observations can separate two descriptions faithful to the same data — the fundamentality question is answered at the level of (S, φ), not adjudicated between its projections.)

5.3 The three-level hierarchy made explicit

The substratum gauge group (§4) provides the structural framework for the synthesis claim at the level of the two gauge-bearing projections (the arrow-of-time projection is developed in §5.4 and has no gauge-hierarchy analog). At the substratum level, the only object is the bijection and its gauge group $\mathcal{G}_{\text{sub}}$. The trace-out projects this onto two derived structures simultaneously: the emergent quantum field theory at the visible-sector level, with the Standard Model gauge group as the commutant of the coupling matrix ([SM, §4.4], Theorem 5) and the Standard Model representations as the cubic decomposition of the link directions ([SM, §4.6], Theorem 7); and the classical horizon thermodynamics at the boundary level, with the metric, the surface gravity, the entropy, and the temperature determined by the partition geometry and the area-law theorem ([GR, §3]). Both derived structures inherit the framework's structural inputs from the substratum, and both are exact rather than approximate at the lattice level.

The three-level gauge hierarchy makes this exact. At Level 3 (the substratum), $\mathcal{G}_{\text{sub}}$ acts on (S, φ) before any trace-out and includes transformations with no analog in the emergent description (state relabeling, alphabet change under

the alphabet-size hypothesis, deep-sector enlargement, graph isomorphism up to statistical isotropy). At Level 2 (the emergent QFT), the Standard Model gauge group acts on the emergent fields as the commutant of the coupling matrix, and is the image of $\mathcal{G}_{\text{sub}}$ restricted to transformations that permute internal components within the eigenspaces of M. At Level 1 (the emergent Hamiltonian), the diagonal-unitary D-gauge acts on the emergent Hamiltonian within the emergent QM and preserves transition probabilities. The complete shift/rephasing/antiunitary classification is conditional on H-BG and the other [GR §3.3] hypotheses. Each level is contained in the one above, and the trace-out projects each onto the next.

The boundary-thermodynamic projection that produces GR is a parallel structure at Level 2 — but acting on the *partition* rather than on the internal field content. Where the emergent QFT is the trace-out's effect on the bulk dynamics, the classical horizon thermodynamics is the trace-out's effect on the partition geometry itself. Both are at Level 2. Both descend from Level 3. The framework's claim is that the SM gauge group and Einstein's equations are not two independent theoretical inputs but two co-derived structures, both consequences of the same underlying (S, φ) and the same trace-out under the same conditions C1–C4; on the gravitational side the covariant field equations are the open Level-G3 step ([GR, §8.7]).

5.4 The arrow of time as a third projection

After Main §4.6 (structural observer-selection theorem), the arrow of time joins quantum mechanics and general relativity as a third emergent structure that the framework derives from the same underlying (S, φ) and the same trace-out machinery. Before §4.6, the arrow of time was treated as external to the derivation chain — a past hypothesis or an anthropic selection, imposed on the framework rather than produced by it. With §4.6 in place, the observer's confinement to the non-equilibrium phase of (S, φ) is a structural fact, and every arrow the observer sees descends from this confinement together with the horizon growth that the accelerating phase produces. The third projection has the same form as the first two: a feature of how the embedded observer sees (S, φ) from inside, rather than a feature of (S, φ) itself.

A distinction should be kept in view here. The *existence* of the temporal domain — that differentiation recurs, that there is more than one moment at all — is not a projection of (S, φ) but the framework's second foundational axiom (the recurrence axiom of the two-axiom observation base; see [Main §1.2] and Part I of the companion paper *Physics Modulo Gauge*). What is a projection, and what this section derives, is the *arrow* — the directedness of time within that domain. The framework derives the arrow; it posits the existence of the domain. Conflating the two would overstate what the projection delivers: §5.4 produces directedness, not the bare existence of succession.

Horizon growth as the structural clock. The framework's primary arrow is

the cosmological horizon entropy $S_{\mathrm{dS}}(t) = A(t)/(4l_p^2)$, monotonically increasing along any observer worldline in the Λ-dominated phase. The monotonicity is a feature of the causal partition rather than of any local frame: every observer in the accelerating phase agrees that S_{dS} increases, just as every observer in a Schwarzschild geometry agrees on the exterior direction. The horizon clock is observer-independent, continuous, and monotonic for as long as $\ddot{a} > 0$, which the framework identifies as the regime of interest for the derivations of [GR]. At the substrate level, (S, φ) has no preferred direction: φ and φ^{-1} are equally valid as forward-time maps. At the observer level, the horizon clock picks out a direction, and all other arrows align with it.

The layered cascade. Four derived arrows align with the horizon clock through a structural cascade. *Cosmological:* $S_{\mathrm{dS}}(t)$ monotonic in the Λ-dominated phase, as above. *Thermodynamic:* via Jacobson's coupling $\rho_s = \rho_{\mathrm{crit}}$ ([GR, §3]), the entropy of any horizon-bounded system increases with the observer's clock — the thermodynamic arrow descends from the structural arrow without an independent postulate. *Memory:* C2 (memory persistence) implies the observer's hidden sector preserves correlations forward along the clock rather than backward, because the coarse-grained mixing time of the hidden-sector dynamics is uniform in the two directions but the observer's measurement protocol commits to one. *Measurement:* the observer records "past" outcomes and prepares "future" states — an asymmetry inherited from the memory layer, not added on top of it. The chain grounds in one structural fact (horizon growth) and one structural constraint (C1–C4 observer selection via Main §4.6). No additional axiom is needed.

What the framework derives, and the directional-language subtlety. The framework's quantitative content at the level of Main §4.6 is *symmetric* in time direction — the spectral-gap bound $\tau_B(V) \leq \tau_{\mathrm{mix}}(\Sigma_{\mathrm{loc}}(V)) + O(\epsilon)$ applies equally in both directions along any observer worldline. What is *imposed* by the observer, not derived, is the convention of which direction along the horizon-clock parameter to call "future." The two directions are gauge-equivalent at the substrate level ($\varphi \leftrightarrow \varphi^{-1}$), and the observer's choice of convention is the residual freedom after all physical content has been extracted. The framework therefore derives the existence of a monotonic parameter along observer worldlines, not a substrate-level arrow of time. This distinction answers Ellis's objection (a bijective substratum cannot carry an arrow of time) by agreeing with the objection's premise while deriving what the observer actually sees from the structure the substratum does have.

Scope note: causal topology as load-bearing. The memory-arrow step uses "the observer's worldline has a causal topology" as the carrier of the asymmetry. This is not a new axiom — [Main, §1.3] already assumes a causal topology on the partition — but it is load-bearing for the cascade and is noted explicitly here rather than treated as obvious. A framework in which the observer's worldline had no causal topology could not run the memory-arrow step, and the cascade would halt at the thermodynamic layer.

Scope note: the arrow derivation is conditional on the ordering of moments. The cascade derives a *direction* along the observer's worldline. It presupposes that the worldline's moments are *ordered* — that they form a sequence — since a monotonic parameter is monotonic with respect to an ordering. This ordering is distinct from the bare existence of more than one moment: the framework's foundational treatment (Part I of the companion paper *Physics Modulo Gauge*) axiomatizes the existence of a temporal domain but treats the *ordering* of its moments as a separate question, which it leaves open. The arrow derivation of this section is therefore conditional: it derives the direction *given* an ordering, and inherits the open status of the ordering question. What this section establishes is the conditional — ordered moments yield a derived monotonic parameter and an observer-imposed direction-convention — not an unconditional derivation of temporal directedness from the substratum alone. The causal topology of the worldline reduces to this ordering together with the dynamics φ (it introduces no commitment beyond the ordering), so the arrow's single open dependency is the ordering question and nothing further.

CP violation as the residual Ellis-class objection. The layered cascade closes the Ellis-axis objection for the arrow of time but does not automatically close the related question of CP violation. The substratum is T-symmetric (load-bearing for the reciprocity of visible transition counts, [SM, §5]; it does not by itself fix $\bar{\theta}$), and observed CP violation in CKM and kaon/B systems is a flavor-basis feature of the specific bijection rather than a consequence of the cascade. The framework treats flavor-sector CP phases as solution-specific inputs — see [SM, §5.4] for the scope statement and §6.4 "what does not falsify" for the commitment that the framework makes no prediction of δ_{CKM}, δ_{PMNS}, or the Wolfenstein (ρ, η).

5.5 What the synthesis does and does not claim

The synthesis claim is structural and architectural rather than empirical. It does not claim that the framework predicts gravitational phenomena that conventional QFT plus GR cannot reproduce — most of [GR]'s predictions (the BH area law, the basic properties of horizon thermodynamics, the broad shape of dark energy phenomenology) are also recovered by other approaches. The framework's empirical content is concentrated in [SM] (the SM derivation, twenty-two quantitative predictions) and in the specific GR-side predictions of [GR] (the Type II running-vacuum channel with $|\nu| \sim 10^{-32}$, MOND $a_0 = cH/6$ with its dimensional factor imported, dark sector concordance), which collectively distinguish the framework from standard ΛCDM plus the Standard Model.

What the synthesis claim *does* establish is that those two papers are not independent. The same construction that produces the SM in [SM] produces the gravitational sector in [GR], and the same trace-out that gives the SM gauge group as the commutant of the coupling matrix ([SM, §4.4], Theorem 5) gives the BH entropy as the boundary mode count ([GR, §5]). This is not a claim about new gravitational phenomena. It is a claim about the structural relation-

ship between two derivations that, in the conventional picture, are independent and incommensurable — and that, in the present framework, are derived from the same object by the same procedure under the same conditions.

This matters for two reasons. First, it removes the QM-GR unification problem from the active research agenda by dissolving rather than solving it: the problem was based on a category error, and the category error is fixed by the substratum-level construction developed here. Second, it provides an explanation for why the SM has the gauge group it does — namely, that the SM gauge group is the visible-sector shadow of the substratum gauge group, with no choice of model and no landscape of alternatives to select among. The SM is not one possibility among many but the unique consequence of the framework's structural inputs.

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6. Discussion

The reconstruction theorem identifies the question "is mathematics describing reality, or is it reality?" as gauge in the precise sense established by §4 — provably undecidable by any measurement an embedded observer can perform, and therefore empty for physics — without foreclosing the question in mathematical foundations or other modes of inquiry not bound by embedded physical observation. It reframes Wigner's puzzle of the unreasonable effectiveness of mathematics as a theorem rather than a mystery. The ontological hierarchy makes explicit that space, time, matter, and energy are derived rather than fundamental concepts. And the measurement problem dissolves once the wave function is recognized as a derived object rather than a component of the underlying reality. The pre-registration of falsification conditions, finally, is the empirical counterpart of the methodological discipline developed in §§3–5: a framework that derives rather than posits must be willing to specify what would invalidate the derivation.

6.1 Ontic structural realism on the equivalence class

The metaphysical commitment introduced in §1.1 can now be stated with the full technical machinery in place. The framework's theorem-level structural-realist commitment applies first to the **local reconstructed residue modulo the stated substratum-gauge freedoms**: Theorem 23 uniquely fixes that local lattice/gauge residue under its stated conditions. H-Bell separately conditions whether a measurement-independent Bell-violating composite can coexist with it, and no theorem here uniquely fixes the Bell-inclusive full completion. Realism about a particular full Bell-complete bijection therefore outruns the reconstruction result.

This is the strongest claim the framework's theorems license at this level without overclaiming. Two alternative framings merit explicit comparison.

Specific-substratum realism — the stronger commitment that a particular bi-

jection φ is the one our universe is in — is forbidden by Theorem 24. The generators of $\mathcal{G}_{\text{sub}}$ relate representatives that produce identical observables, so no measurement can discriminate among them; specific-substratum realism commits to facts the framework's own gauge structure classifies as unobservable.

Pure explanatory economy — the weaker position that the framework is simply more economical than standard QM (fewer postulates, more derivations) without ontic commitment — undersells what Theorem 23 establishes about the **local reconstructed residue**. Under the theorem's stated conditions that residue is not one local possibility among many but the uniquely fixed local lattice/gauge structure modulo gauge. The Bell-inclusive completion is a separate underdetermined layer, so explanatory economy follows from local-sector reconstruction without implying full-substratum uniqueness.

The wave function, Hilbert space, emergent gauge group, and metric are features of representations — of the emergent description of a specific partition of a specific substratum — not of the equivalence class itself. At the structural level there is only the bijection and its symmetries; no quantum, no classical, no metric, no wave function. This is the structural reason the QM-GR incompatibility dissolves: both descriptions are emergent from the same structural object at different levels of projection (§5). Asking whether the wave function is "real" is then asking about a feature of the representation, and the question has a two-level answer — the wave function is real at the emergent level, gauge at the substratum level.

The question "is mathematics describing reality, or is it reality?" is, on this reading, gauge for the embedded physical observer **within the reconstructed local observable structure**: the local residue is determinate modulo the stated gauge freedoms, while Bell-completion data are not uniquely reconstructed. The metaphysical attitude one takes toward either description is information the framework does not encode in accessible observables, and therefore no measurement an embedded observer can perform can determine it. The framework does not claim the question is absolutely meaningless. Whether it has substantive content in mathematical foundations or other modes of inquiry not bound by embedded physical observation is a separate matter the framework does not address. Wigner's "unreasonable effectiveness of mathematics" becomes a theorem modulo the structural assumptions rather than a mystery: physics is structural, mathematics is the language of structure, and the effectiveness is the reconstruction theorem.

Multi-level structural realism. The structural-realism commitment is not single-level. [Structure] establishes that the framework operates at multiple levels of structural realism with different content at each level. The hierarchy is two-dimensional: an observation-hierarchy axis (vertical depth — from foundational to specific) and a gauge-hierarchy axis (horizontal breadth — from narrow to wide gauge equivalence). The combination produces a structure of realisms operating at multiple intersections.

At the broadest level — universality-class equivalence across structural classes (Level G4 in [Structure §2]) operating at the partial-trace observational features sub-class (Level C in [Structure §2]) — what is real is the algebra-channel structure of partial-trace observation: availability of a fixed-basis Born-form representation, channel-level unitarity in that representation, the non-Markovian marginal in the readback sector, and the commutant gauge-invariance pattern. These are representation/observation features shared across the stated observer-admitting class; the claim does not independently select the quadratic exponent or close the coherent local operational lift.

At intermediate levels, realism applies to the SM gauge group (forced uniquely in OI but shared with any SM-reproducing universality class) and the algebra-channel structure within the SM-reproducing sub-class.

At the most class-specific theorem-supported level — the **local reconstructed residue modulo $\mathcal{G}_{\mathbf{sub}}$** within OI's structural class (Level G3 in [Structure §2]) operating at OI's specific universality-class representative (Level D in [Structure §2]) — what is real is the local lattice/gauge structure Theorem 23 fixes. A stronger realism about the full Bell-complete substratum is additional and presently underdetermined. OI's specific predictions (Cabibbo angle $1/(\pi\sqrt{2})$, Koide ratio 2/3, dark-sector phenomenology) are real at this level, distinguishing OI's universality-class representative from alternatives within the same partial-trace-features sub-class.

These are not competing realisms; they are realism at different levels of the hierarchy. The framework commits to all of them simultaneously, with content at each level being class-specific (more specific at deeper levels) or class-universal (more universal at higher levels). What is real at the universality-class level is what observation extracts from any partial-trace operation; at the deepest theorem-supported level it is the specific **local structural residue** identified by the reconstruction theorem, with Bell-completion structure left open.

This refinement does not weaken the substratum-class realism articulated above; it situates it within a broader hierarchical structure. The framework's empirical content (twenty-two quantitative predictions, specific cubic-lattice substratum, specific gauge group derivation) is concentrated at the deepest level of the hierarchy — Level D × Level G3 — which is where the reconstruction theorem and substratum gauge group operate. The broader levels of the hierarchy are where the framework connects to other unification programs and where universality-class structural realism applies. The full articulation of the hierarchical structure is in [Structure §2]; the substratum-class realism developed in this paper is one specific level within that broader structure.

6.2 The ontological hierarchy

The triple (S, φ, V) generates every concept in fundamental physics, not as independent substances but as different aspects of the same structure. **Space** is the coupling structure of φ — the graph G_φ determined by which degrees

of freedom affect which others ([SM, §2.4]). **Matter** is the state — localized patterns that propagate through the coupling graph. **Energy** is the rate of change under iteration. **Time** is the iteration itself. **Quantum mechanics** is the observer's compressed description of the visible sector. **General relativity** is the thermodynamic limit of the coupling structure. **Conservation laws** are emergent: energy conservation (Noether) is what information conservation (bijectivity) looks like in the emergent quantum description. None of these are independent entities; they are descriptions of (S, φ, V) at different scales.

6.3 The measurement problem

On the structural reading, the measurement problem is dissolved. The wave function is not a component of (S, φ, V) — it is a derived object. Since it is derived, not fundamental, asking "does it collapse?" is asking about the behavior of a compression artifact. In the double-slit experiment, the particle traverses a single slit in the deterministic substratum. In Wigner's friend, the Friend has a definite outcome; Wigner's superposition reflects his epistemic deficit.

Branching is forbidden by the rigidity of φ. A fixed bijection on a finite set has exactly one trajectory from any initial state. There is no point at which the trajectory splits. The appearance of branching in the emergent quantum description reflects the observer's uncertainty about which trajectory they are on (because they cannot see the hidden sector), not a physical splitting of worlds.

Bell correlations force a sharper distinction. For any **finite-range Bell-relevant** substratum, the dependency graph gives an exact causal cone; if a response-complete deterministic completion also uses one setting-independent pre-setting ensemble, Bell factorization follows and CHSH is capped at 2. The framework therefore does not obtain quantum Bell violation from screening plus indivisibility. It keeps measurement independence and takes ontic parameter dependence: Bell-violating substrata have Bell-relevant coupling range exceeding the wings' separation, while the averaged operational statistics remain no-signaling [Main §3.3]. The stochastic causal-local indivisible Tsirelson theorem is retained only at that weaker stochastic layer.

6.4 Pre-registered falsification conditions

The framework's structural content — the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$, the emergent Standard Model structure, and the co-derivation of quantum mechanics and general relativity — makes specific commitments whose future observational status can now be stated explicitly. This subsection pre-registers the conditions under which the framework would be falsified, separated from properties of the specific bijection φ that the framework does not claim to determine.

The distinction matters for two reasons. First, an explicitly pre-registered falsification schedule is the empirical counterpart of the methodological discipline

developed in §§3–5: a framework that derives rather than posits must be willing to specify what would invalidate the derivation. Second, pre-registration separates structural commitments — which the framework stands or falls on — from solution-specific properties that lie outside its scope, and which continued openness on does not undermine the framework.

Class A: Structural theorems. The following are claims about the equivalence class itself and would falsify the framework's reconstruction if violated.

- *Strong CP.* The construction does not fix $\bar{\theta}$ (SM §5.3, Theorem 21: T-invariance leaves $\theta \in \{0, \pi\}$ and does not make the emergent Hamiltonian T-invariant), so the framework makes no commitment on the neutron electric dipole moment; reciprocity leaves the Jarlskog invariant free, and a mechanism making Y_u and Y_d simultaneously real would force $J = 0$ — the falsification control on H-det proposals (SM §5.3–5.5); a measured $\bar{\theta} \neq 0$ would constrain mechanisms proposed for H-det, not the T-symmetry of the substratum, which is compatible with any value of $\bar{\theta}$.
- *Gauge group SU(3) × SU(2) × U(1) exactly.* Any observation of a stable beyond-Standard-Model gauge structure — an additional Z' interpretable as a new gauge boson rather than a composite resonance, stable exotic matter transforming under a novel gauge group, or any confirmed extension of the visible-sector gauge content — falsifies the cubic-group decomposition of SM §4.6.
- *Three fermion generations.* Any observation of a stable fourth-generation fermion with Standard Model quantum numbers falsifies the taste-reduction derivation (SM §4.7). Non-stable exotic states consistent with composite or resonance interpretation do not.
- *Majorana neutrinos with normal ordering.* Inverted mass ordering, confirmed to high significance, falsifies the taste-breaking derivation of the neutrino mass matrix (SM §8.4). Confirmed Dirac nature of the neutrino (e.g., a null result for neutrinoless double-beta decay at the sensitivity that would require Majorana masses well below the framework's prediction) falsifies the prediction that no right-handed neutrino exists in the spectrum.
- *Bell/Tsirelson test.* For the cited **stochastic causal-local indivisible** sector, $S_{\mathrm{CHSH}} \leq 2\sqrt{2}$. At the deterministic level, measurement independence plus ontic parameter independence instead gives the Bell-local ceiling 2, so the framework's Bell-violating branch is ontically parameter-dependent [Main §3.3]. A confirmed loophole-free excess above $2\sqrt{2}$ would falsify standard quantum mechanics and any claim that the operational lift lands in the standard quantum Bell set; it would not by itself falsify P-indivisibility. Bare C1–C4 or graph locality do not enforce Tsirelson.
- *Leading Lorentz-invariance-violation coefficient.* The cubic-lattice dispersion relation ([SM, §4.4] Theorem 4, generalized to d = 3: $\cos(\omega\epsilon) = \frac{1}{3}\sum_{i=1}^{3}\cos(k_i\epsilon)$ for a massless mode) yields standard emergent dispersion $\omega = |k|$ at leading order. Series expansion gives the first correction as a

direction-dependent quadratic term $\omega^2 = k^2[1 - 2\delta(\hat{k})(k\epsilon)^2 + O((k\epsilon)^4)]$, with $\delta(\hat{k}) = \frac{1}{24}\sum_i \hat{k}_i^4 - \frac{1}{72}$ evaluated on the rigid $\mathbb{Z}^3$ representative. The rigid-lattice anisotropy (values $\delta_{[100]} = 1/36$, $\delta_{[110]} = 1/144$, $\delta_{[111]} = 0$ — the dispersion is exactly linear along the body diagonals, $\omega = k/\sqrt{3}$ to all orders) is a *physical, direction-dependent* prediction. Its absolute orientation in space is gauge: generator (iv) of §4 gauges the rigid orientation of the cubic representative. The anisotropy *pattern* — its fourfold $\ell = 4$ cubic-harmonic structure and the direction-differences $\delta_{[100]} - \delta_{[111]}$ — is gauge-invariant, since generator (iv) preserves the octahedral structure and does not identify $\mathbb{Z}^3$ with a generic isotropic graph (which lacks the octahedral structure the gauge group requires, [SM §4.5–4.6]). The orientation-averaged coefficient $\langle\delta\rangle = 1/90$ is a direction-summary of this anisotropy, and the framework predicts the emergent dispersion

$$\omega^2 = k^2\left[1 - \frac{(k\epsilon)^2}{45} + O\left((k\epsilon)^4\right)\right] = k^2\left[1 - \frac{4\sigma}{45}\left(\frac{E}{M_{\mathrm{Pl}}}\right)^2 + O\left(\left(\frac{E}{M_{\mathrm{Pl}}}\right)^4\right)\right]$$

after using $\epsilon = 2\sqrt{\sigma}\, l_p$ and $k\epsilon = 2\sqrt{\sigma}\, E/M_{\mathrm{Pl}}$; the coefficient of $(E/M_{\mathrm{Pl}})^2$ is $4/45$ at $\sigma = 1$. Three structural features: (i) *subluminal*, since $\delta(\hat{k}) \geq 0$ everywhere on the sphere ($\langle\delta\rangle > 0$ is invariant under orientation-averaging); (ii) *no linear-in-E/M_{Pl} term*, because the even-parity cosines in the dispersion admit no $\mathcal{O}(k^3)$ contribution — any demonstration of linear LIV at any scale therefore falsifies the framework's substrate-level commitment to the cubic-lattice wave equation; (iii) *specific lattice coefficient* $1/45$ of $(k\epsilon)^2$, fixed by the cubic-lattice coordination and isotropy-gauge average, not tunable. Features (i)–(iii) and the anisotropy pattern are the Class A content, at tree level under the stated assumptions; the directional ratios $\delta_{[100]} : \delta_{[110]} : \delta_{[111]} = 4 : 1 : 0$ are independent of σ. The physical amplitude, the coefficient $4\sigma/45$ of $(E/M_{\mathrm{Pl}})^2$, carries the undetermined entropy σ per boundary mode and is not a parameter-free numerical prediction: a measurement of it would constrain σ rather than falsify the coefficient independently. Protection of the tree-level coefficient in the interacting theory is open ([SM §3.1]). Current quadratic-LIV bounds from GRB time-of-flight reach only $E_{\mathrm{QG}} \gtrsim 10^6$ GeV (roughly thirteen orders of magnitude below M_{Pl}), so the amplitude $4\sigma/45$ is not yet near-term testable at the magnitude level; linear-LIV tests, by contrast, are already at sensitivities where any positive detection falsifies.

Class B: Parameter-free retrodictions. The framework produces a set of numerical predictions that follow from the structural content with no adjustable parameters beyond a small set of acknowledged empirical inputs (m_s for the overall fermion mass scale, m_t for the Higgs RGE running, η for the Jarlskog phase, Q_{down} for the down-sector Koide normalization). These predictions are stratified into five sub-classes reflecting their structural status:

Class B-S (structural). One prediction is a Layer 1 structural retrodiction with

no open link specific to the entry; like every Class B entry it rests on the matter-sector hypotheses H-spin' and H-χ':

- The Wolfenstein parameter $A = \sqrt{2/3}$ ([SM §7.1]).

A global-fit A outside $\sqrt{2/3}\,(1\pm\lambda^2)$ would strain the leading-order identification, while the protected $\sin\theta_{13} = A^2\lambda$ comparison tests the geometry directly ([SM §7.1]).

Class B-C (conditional structural). Two predictions are Layer 1 retrodictions with no fitted input whose derivation chain has one named open link:

- The Cabibbo angle $\lambda = 1/(\pi\sqrt{2})$ ([SM §7.1]) — the chirality verification of the taste-changing vertex's spinor structure is closed via Mason et al. (HPQCD, hep-lat/0209152), establishing that taste-changing transitions in staggered quarks have spinor structure given by a combination of γ_μ and $\gamma_{5\mu}$ (both chirality-preserving); the identification leaves the emergent-theory dressing open, and the prediction is conditional on it.
- The Koide angle $\theta_0 = C_2/d^2 = 2/9$ ([SM §7.2]) — matched to observation (0.001%) and the cleanest structural derivation (cubic-group quadratic Casimir over bandwidth squared, with the §7.2 uniqueness table enumerating ten alternative dimensionless ratios that fail to match); it is conditional with the Koide amplitude, whose value $A^2 = C_2$ is not yet derived from the dynamics ([SM §7.2]).

A precision-upgrade violation would falsify the identification or the closure on which the entry is conditional.

Class B-L (layered conditional, pending open derivations). Six predictions (the two PMNS angles counted separately) follow from the structural form combined with one Layer-2 substrate input, one empirical input, or one named condition on the horizon reservoir. Each has an explicit derivational gap whose closure would move the prediction to B-S:

- The mass ratio $m_d/m_s = 1/(2\pi^2)$ ([SM §7.1]), its conversion from the Cabibbo scale running through the Gatto-Sartori-Tonin relation, an external empirical relation whose status the chain inherits.
- The mass ratio $m_u/m_d = \sqrt{\theta_0} = \sqrt{2/9}$ ([SM §7.2]) pending the explicit derivation of the "different channels" mechanism producing the square root, structurally analogous to the open OI-vertex 1-loop derivation for $K = 1/2$ ([SM §7.5]) in being mixed-layer — substratum cubic-group structure combined with electroweak-symmetry-breaking emergent machinery.
- The mass ratio $m_b/m_\tau = 4.28/Z_S$ ([SM §7.5]) pending the explicit OI-vertex 1-loop computation confirming $K = 1/2$, together with four bridge gaps in the SχPT inheritance ([SM §7.5]): the substratum measure reduction under Theorem 2, the 3D MC vs 4D SχPT dimensional bridge, the 3D BZ vs 4D Dirac taste-structure equivalence, and the 3D vs 4D parity identification of chirality. The per-vertex $\cos^2$ symmetrization producing $K = 1/2$ is dimension-independent and structurally robust; the

four bridges are derivation gaps inherited by the other §7 mass-cluster predictions that share the SχPT machinery. The closure path is the two-loop SχPT calculation specializing Panagopoulos–Spanoudes 2017 to the OI cubic-group setting.

- The PMNS predictions $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2)$ and $\sin^2\theta_{23} = 1/2 + 1/(2\pi^2)$ ([SM §7.3]) pending derivation of Cond 2 (the structural relation among A_2 Wilson coefficients) from cubic-lattice Yukawa structure. The reactor angle $\sin^2\theta_{13}$ is independent of Cond 2; it inherits the Cabibbo λ and is conditional with it (class C in [SM §7.6]; not counted separately in Class B).
- The MOND critical acceleration $a_0 = cH/6$ ([GR §7.3]), inheriting the conditions on the de Sitter temperature of [GR §3.2] and the coupling statement $Q = M_B c^2$ that the open G3 map would supply, with the 1/6 dimensional factor imported from Verlinde 2016 (eq. 1.7) as $(d-3)/[(d-2)(d-1)]$ in $d = 4$. Dependent predictions (baryonic Tully-Fisher, crossover radius r_M) inherit B-L status.

Class B-L violations at 3σ on precision upgrade are addressed under the pre-registration commitment via three branches: (a) closure of the open derivation *confirms* the prediction (recoverable from a transient data fluctuation); (b) closure *fails to confirm* the prediction while preserving the framework's structural commitments (recoverable error in the specific derivation, with the structural commitments intact); (c) demonstration that no consistent OI-internal derivation can produce the observed value (fatal, framework-falsifying).

Class B-M (mass chain). Two predictions inherit a single empirical input scale through a structural relation:

- The electron and muon masses m_e and m_μ via the Koide chain from the empirical input m_τ ([SM §7.2]). The prediction is structurally $\theta_0 = 2/9$ with $Q = 2/3$; the absolute scale is set by m_τ.
- The Higgs mass m_H via SM RGE running from the structural boundary condition $\lambda(M_{\rm Pl}) = 0$ ([SM §7.4]) with m_t as the empirical input.

Class B-M predictions falsify under the same precision-upgrade protocol as B-L, but the inheritance is from an empirical input scale rather than from an open derivation.

Class B-R (retrodictions). Two gauge-coupling values are explicitly retrodictions per the §6.3 [SM] parameter count:

- The SU(2) gauge coupling at M_Z ([SM §6, items 11-12 in §7.6]).
- The SU(3) gauge coupling at M_Z.

Three fitted parameters (δ_0, A, B) against three observed couplings produces zero residual by construction; these classifications are honest. Class B-R predictions can never falsify in the precision-upgrade sense — their value is what falsification of the *framework's gauge-sector structural commitments* (universality of δ_0, the resummation form, the structural prediction $1/\alpha_0 = 23.25$ from

the 1-loop staggered VP (ambient coefficient; native for hypercharge only under H-observer-bundle and H-Y-vertex, [SM §6.5]), and the $A \cdot B$ cross-check (≈ 48 under the $1/N^2$ leading-power assumption; inconclusive across finite-size models, [SM §6.2.1])) would test.

Status summary. Of the predictions in Class B: **1 is structural with no open link of its own** (B-S), **2 are conditional structural** (B-C), **6 are layered conditional pending open derivations** (B-L), **2 are mass-chain inheritances** (B-M), and **2 are retrodictions** (B-R). Total 13 predictions in Class B. Their empirical standing is assessed entry by entry: the gauge couplings match by construction, and the quoted precision of a measured comparison does not apply to every entry.

The current status of each is documented in the SM and GR companion papers with experimental references. As experimental precision improves, the discrimination power of the B-S set increases monotonically; closure of the open derivations would move B-C and B-L predictions to B-S over time.

The Bekenstein-Hawking coefficient $S = A/(4l_p^2)$, with the factor 1/4 the ratio $2\pi/(8\pi)$ between the Euclidean KMS period and the Jacobson Einstein-equation prefactor ([GR §5]), is a consistency relation of the horizon calibration and is not counted among them. It is conditional on the infrared detailed-balance condition H-slope on the horizon reservoir together with the horizon and frame conditions of [GR §2, §8.5]; the counting $S = \sigma A/\epsilon^2$, for an entropy σ per boundary mode, is unconditional and the coefficient enters through $\epsilon^2 = 4\sigma\, l_p^2$, from which σ cancels. The closure path is the continuum-generator theorem of [GR §2]. GW250114 confirms the classical area theorem, which the framework preserves; the 1/4 coefficient has no direct empirical test.

Class C: Framework-level commitments. Two broader commitments are implicit in the construction; each is stated with what would contradict it.

- *Finite classical realizability of quantum processes.* The characterization theorem ([Main, §3.4]) establishes that the quantum representation is the description admitted by an embedded observer satisfying C1–C4 on a deterministic substrate with bounded classical hidden-sector memory. At finite accuracy, every finite adaptive quantum experiment has a finite deterministic realization (the gluing and uniform-continuum theorems of [Main, §3.4]), while the exact continuum theory at any fixed finite carrier is excluded by the classical-dimension obstruction. Non-Markovian models are permitted and expected (they are how the hidden-sector correlation persistence manifests). No finite-accuracy experiment contradicts this commitment by itself. A memory-witness experiment would contradict the ontology only given the coherent-preparation and instrument bridge, an outcome-independent rule identifying fundamental open-system processes, and a constraint derived from the completed model on what those processes must show; the last is not established ([Main, §4.5]).
- *The dark sector as entropy-displacement, not as a novel particle content.*

The framework's account of dark energy as boundary-entropy bookkeeping read in the Type II RVM channel, and dark matter as frozen boundary entropy (GR §7.3), is incompatible with the direct detection of a WIMP interpretable as a fundamental particle with Standard-Model-like interactions and no entropy-displacement signature. A confirmed WIMP with such properties falsifies the framework's dark-sector account while leaving the rest of the structure intact — it would require abandoning the §7.3 derivation and either identifying an error in it or accepting that the framework's dark-sector story is wrong even as other derivations survive.

Cosmological commitment. The dark energy sector is subtler than Classes A–C. The framework supplies geometry-dependent dark-energy bookkeeping whose smooth late-time expansion contains an $H^2 - H_0^2$ term — the local form shared by every smooth model — and reads it in the Type II running-vacuum channel; it does not presently derive a distinctive Type II law (GR §7.1). The magnitude is given by the emergent-QFT calculation $|\nu_{\mathrm{QFT}}| \sim (M_{\mathrm{SM}}/M_{\mathrm{Pl}})^2 \sim 10^{-32}$ (conditional on the GR §7.3 split, not yet established at theorem level), observationally indistinguishable from zero. Substratum contributions beyond this are constrained by the boundary mode count of §3-§5, which catalogues modes by spatial location alone (one mode per ϵ^2 cell, no inherent frequency assignment — a count forced by the link-direction structure of the $K = 2d$ components, [GR §3.1]); the natural 2D distribution concentrates them near the UV cutoff, contributing well below $|\nu_{\mathrm{QFT}}|$. The framework's commitment is therefore the Type II reading, with magnitude $|\nu| \approx |\nu_{\mathrm{QFT}}|$ (conditional, per GR §7.1), and the Bertini et al. (2025) measurement $\nu = -(2.5 \pm 1.3) \times 10^{-4}$ consistent with this expectation at 2σ, where Bertini is itself consistent with zero. The falsification band is correspondingly stratified: a DESI Year 5 measurement of $|\nu| \gtrsim 10^{-4}$ at high significance would require additional substratum structure (log-uniform mode-per-decade distribution at the cosmological horizon) not currently in §3-§5; a measurement of ν with high-significance positive sign would falsify the Type II functional form itself, which is a structural commitment rather than a magnitude one. The first outcome is a refinement within the current bracket; the second is a structural falsification.

What does not falsify. The framework makes no commitment to the specific values of solution-specific quantities, and their future measurement does not bear on the framework's structural status. These include: the absolute scale of fermion masses (the choice of m_s as empirical input is acknowledged in SM §7.6); the CP-violating phases in CKM and PMNS (solution-specific properties set by the basis mismatch between up-type and down-type Yukawa mass eigenstates; $\bar{\theta}$ is not fixed by the construction and reciprocal transition counts do not force the Jarlskog invariant to vanish [SM §5.3–5.4], so the phases are free solution-level data); the numerical value of the Type II ν coefficient within the bracketed range above; initial conditions at any cosmic time (the framework derives the structure from which evolution proceeds, not the state at t_0); and the specific microstate φ within the equivalence class (Theorem 24 classifies this as gauge,

so the question of which representative our universe is in is provably without empirical content). A framework is not falsified by the presence of solution-specific inputs; it is falsified by failure of its structural commitments. The Class A–C list, together with the stratified dark-energy band, is the complete structural commitment of the framework as currently written.

The pre-registration commitment. The author commits to treating the Class A conditions, the Class C commitments, and a Class B retrodiction exceeding 3σ on precision upgrade as framework-falsifying rather than as an invitation to post-hoc rescue. In the case of a Class B violation, the framework will be revisited to determine whether the violation reflects a computational error in the specific retrodiction, an incorrect specific-bijection input, or a structural problem with the derivation; the first two are recoverable, the third is fatal. The framework does not reserve the right to add structural postulates after the fact to accommodate disconfirming evidence. Pre-registration of this commitment is itself part of the falsifiability content of the framework.

Regimes of deliberate silence. Three regimes lie outside the framework's predictive scope by construction and are not objects of falsification claims. First, the *pre-horizon early universe* — the regime before a stable causal partition with $\tau_B \sim H^{-1}$ exists — has no observer in the framework's sense ([Main, §1.3]) and no horizon-level structure for the [GR] derivations to act on; the framework's only commitment in this regime is a boundary condition on deeper theories, namely that any successful pre-horizon dynamics must produce a C1–C4-satisfying regime by the time a horizon stabilizes. Second, the *equilibrium phase* of (S, φ) is structurally observer-free by the structural observer-selection theorem ([Main, §4.6]); the framework makes no prediction for this regime because it has no observers to whom a prediction could be addressed. Third, regions *beyond the observer's causal horizon* are part of that observer's hidden sector and are addressed only through boundary effects (the area-law entropy of [GR, §5]); the framework takes no observer-independent position on their internal content. These silences are structural rather than methodological — they follow from the characterization theorem's identification of QM with embedded observation, and from the observer-selection theorem's restriction of observers to the non-equilibrium phase — and stating them explicitly sharpens the scope of the positive commitments above.

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7. Conclusion

The substratum-level results developed in this paper — the reconstruction theorem (Theorem 23) and the substratum gauge group (Theorem 24) — turn the framework's three derivations into a layered single-construction claim. Under M1-T, C2-mixing and Lemma 24.1's completeness step, the reconstruction uniquely fixes the **local propagating lattice/gauge residue** modulo the stated gauge freedoms; M1-B and H-Bell separately condition the existence of

a measurement-independent Bell-violating completion, whose uniqueness is not established. The local reconstruction carries with the Standard Model gauge group as a forced retrodiction, and the anomaly-free hypercharge assignment forced once the observed family pattern is given (fermion embedding closed via the link-carrier construction, with the candidate-sector count locked at three by coupling-degree minimality (three physical generations only under H-spin', [SM §4.7]) jointly with the condensate-stabilizer accounting). The substratum gauge group identifies the kernel of this inverse map and shows that the Standard Model gauge group is its visible-sector shadow. Together, these two results establish that the framework's emergence of quantum mechanics [Main], its derivation of the Standard Model [SM], and its derivation of the gravitational sector [GR] are not three independent applications of the same trace-out machinery but three projections of one object — the bijection (S, φ) — viewed at different levels of description. *Established* here carries the scope of Theorem 23: the local propagating lattice/gauge residue is what the reconstruction fixes, modulo the stated gauge freedoms. A Bell-violating completion of the same object exists only conditional on H-Bell, and its uniqueness is not established, so the single-object claim is layered rather than total — it is a claim about the reconstructed local residue together with a conditional Bell sector, not about a uniquely selected $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$ entire.

The synthesis claim that follows is not a unification of quantum mechanics and general relativity in the traditional sense. The traditional unification problem treats the two theories as competing accounts of the same regime and asks how to merge them. The present framework treats them as projections of the same construction onto two different substructures — the visible-sector trace-out and the boundary-thermodynamic limit — and the question of how to merge them is dissolved as a category error rather than answered as a technical problem. The result is not a theory of everything but a structural account of why the apparent incompatibility between quantum mechanics and general relativity has been so resistant to resolution: there is nothing to resolve, because the two are not in competition. They are co-derived from the same object by the same procedure under the same conditions, and the framework developed across this four-paper synthesis ([Main], [SM], [GR], [Substratum]) makes this exact rather than approximate.

Three sets of open problems remain. Neither these results nor this four-paper synthesis require the framework to be the final theory of physics: the substratum may itself be an effective description of something deeper, and the trace-out may have corrections beyond the leading-order results developed here. First, the *specific Bell-inclusive completion* that describes our universe is not uniquely determined by the framework. Theorem 23 fixes the local lattice/gauge residue modulo its stated gauge freedoms, while H-Bell leaves the preparation-indexed Bell-composite rule underdetermined. The framework predicts the structural features of the Standard Model and the gravitational sector but not the specific values of (for example) the lightest fermion mass or the specific numerical value of the Type II RVM coefficient ν beyond its dimensional order of magnitude and

sign. These are properties of the particular bijection, analogous to the mass of the sun in general relativity. Second, the *trans-Planckian regime* — the regime in which the lattice spacing $\epsilon = 2\sqrt{\sigma}\, l_p$ is not small compared to the relevant length scale — lies outside the framework's leading-order results. The framework presents the lattice as fundamental, not as a regulator approximating a continuum theory, and so does not need a continuum limit in the standard sense; but the corrections to the leading-order results in the regime where the partition geometry varies on lattice scales are an open question. Third, the *initial conditions* — the specific configuration of the bijection at any given time — are likewise not determined by the framework. The framework establishes which structures emerge from the bijection but not which microstate the universe is currently in.

These open problems are framed within the construction rather than against it. Each is sharply formulated and admits a definite (if presently unanswered) form. The progress reported here is that the structural relationship between quantum mechanics and general relativity — the central open problem of fundamental physics for nearly a century — is fixed by the construction developed across this four-paper synthesis, and the remaining open problems are problems within the construction rather than problems with it.

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